

Sid Mathur's Thematic ETF Screener – Daily Report

2026-06-25

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Action Center

Run date: 2026-06-25 **Universe:** 660 ETFs / 26 clusters

Full PC1: FLAT **Clusters long:** 8/26 **ETFs long:** 214/660 (32%)

Today's events (cluster level)

spread_buy = 5dma – 50dma; spread_sell = 5dma – 20dma (price units). A BUY fired today iff spread_buy crossed above 0; a SELL iff spread_sell crossed below the cushioned threshold.

SELL (1)

cluster	theme	spread_sell	p_h5	nss
cluster_5	U.S. Large Cap Equity	-1.282	0.97	-0.761

Imminent (cluster level)

Ranked by short-horizon NSS (h=1 day) so candidates that could trigger *tomorrow* surface above those that need several days; p_h5 is the 5-day probability for context. **spread_buy = 5dma – 50dma, spread_sell = 5dma – 20dma** (price units; closer to 0 = closer to crossover). **Imminent BUYs are restricted to clusters whose 252d trend filter currently approves** – otherwise the cross would be blocked at entry, so the imminence is moot.

Imminent BUY (top 5 by h=1 NSS, filter-approved)

cluster	theme	spread_buy	p_h1	p_h5	nss_h1	nss_h5
cluster_14	Financials	3.458	1.000	1.00	1.000	1.00
cluster_1	U.S. Mid Cap Equity	3.842	1.000	1.00	1.000	0.99
cluster_22	Japan Equity	6.900	1.000	1.00	1.000	0.93
cluster_24	Cash / Ultra-Short Fixed Income	0.075	0.985	0.58	0.857	0.23
cluster_9	Emerging Markets Equity	3.249	0.998	0.84	0.685	0.44

Imminent SELL (top 4 by h=1 NSS)

cluster	theme	spread_sell	p_h1	p_h5	nss_h1	nss_h5
cluster_19	Broad Commodities	-10.221	1.000	1.00	-1.000	-1.00
cluster_25	International Fixed Income	-0.843	1.000	1.00	-1.000	-1.00
cluster_11	High Yield Bond	-0.046	0.591	0.42	-0.495	-0.23
cluster_18	Consumer Staples	0.389	0.157	0.28	-0.012	0.00

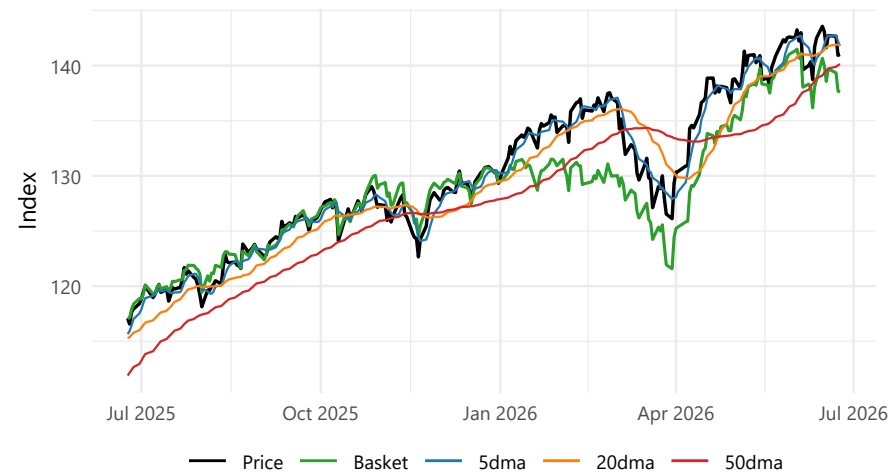
Full Sample PC1

Figure 1: Full sample PC1 portfolio price with short / sell / buy moving averages and replication-basket overlay (last 12 months). Basket rescaled to start at PC1's value at the left edge of the visible window.

Replication basket: VOO basket cor: 0.94 sticky

Current state: FLAT

5dma: 141.790 20dma: 141.836 50dma: 140.137

Spread vs 50dma: +1.653 Spread vs 20dma: -0.047

P(buy crossover within 5 days): 79.4% P(sell): 59.7%

Position Sheet

Replicate each currently-long cluster via its 1-5 member basket. Weights are NNLS (non-negative least squares) – long-only and tradeable – and sum to ~1 per cluster for unit-variance replication. Scale each cluster's basket to your target cluster allocation.

Active positions (long clusters)

ticker	cluster	theme	nss	days	return	pos_wt	cor
VHT	cluster_12	Healthcare	1.000	33	6.2%	10.28%	0.93
MUB	cluster_16	Municipal Bond	0.962	18	0.4%	18.09%	0.95
BND	cluster_4	Aggregate Fixed Income	0.805	4	0.5%	20.14%	0.99
VTV	cluster_2	U.S. Large Cap Value	0.224	50	4.7%	16.56%	0.98
XLP	cluster_18	Consumer Staples	-0.004	6	0.2%	7.44%	0.99
XLU	cluster_18	Consumer Staples	-0.004	6	0.2%	4.61%	0.99
HYG	cluster_11	High Yield Bond	-0.231	4	-0.2%	18.33%	0.95
BWX	cluster_25	International Fixed Income	-1.000	17	-2.6%	15.01%	0.93
PDBC	cluster_19	Broad Commodities	-1.000	14	-14.5%	6.47%	0.98

Watchlist (imminent-buy clusters, $p > 80\%$, not yet long)

ticker	cluster	theme	p_h5	nss	weight	cor
XLF	cluster_14	Financials	1.00	1.000	1.097	0.96
IJH	cluster_1	U.S. Mid Cap Equity	1.00	0.990	0.994	0.99
EWJ	cluster_22	Japan Equity	1.00	0.935	0.909	0.97
QQQ	cluster_7	U.S. Large Cap Growth	0.95	0.249	1.023	0.98
VXF	cluster_6	U.S. Extended Market Equity	0.88	0.292	1.013	0.97
IEMG	cluster_9	Emerging Markets Equity	0.84	0.438	0.923	0.99

Cluster Overview

name is derived from the basket members' categories (e.g. "Technology" or "Tech + Financials"). change shows the LONG/FLAT shift vs the prior run: →LONG or →FLAT for a state flip, new if this cluster had no match in yesterday's bundle, shifted if the cluster's identity changed (top-3 members by |PC1 weight| no longer match), empty if state is unchanged. dNSS is today's NSS minus the prior run's.

	id	name	category	n	signal	change	days	return	nss	dNSS
12	cluster_12	Healthcare	Healthcare	17	LONG		33	6.2%	1.000	+0.002
14	cluster_14	Financials	Financials	14	FLAT				1.000	-0.000
1	cluster_1	U.S. Mid Cap Equity	U.S. Mid Cap Equi...	83	FLAT				0.990	-0.003
16	cluster_16	Municipal Bond	Municipal Bond	13	LONG		18	0.4%	0.962	-0.007
22	cluster_22	Japan Equity	Japan Equity	9	FLAT	shifted			0.935	+1.935
4	cluster_4	Aggregate Fixed Income	Aggregate Fixed I...	58	LONG		4	0.5%	0.805	+0.025
9	cluster_9	Emerging Markets Equity	Emerging Markets ...	25	FLAT				0.438	-0.224
6	cluster_6	U.S. Extended Market Equity	U.S. Extended Mar...	41	FLAT				0.292	-0.185
7	cluster_7	U.S. Large Cap Growth	U.S. Large Cap Gr...	39	FLAT	shifted			0.249	-0.151
24	cluster_24	Cash / Ultra-Short Fixed In...	Cash / Ultra-Shor...	9	FLAT				0.230	-0.072
2	cluster_2	U.S. Large Cap Value	U.S. Large Cap Va...	69	LONG		50	4.7%	0.224	-0.325
18	cluster_18	Consumer Staples	Consumer Staples	11	LONG	shifted	6	0.2%	-0.004	+0.016
20	cluster_20	India Equity	India Equity	10	FLAT	shifted			-0.033	+0.952
15	cluster_15	Real Estate	Real Estate	14	FLAT				-0.221	-0.231
11	cluster_11	High Yield Bond	High Yield Bond	21	LONG		4	-0.2%	-0.231	-0.393
26	cluster_26	Leveraged / Inverse	Leveraged / Inver...	2	FLAT	shifted			-0.356	+0.207
3	cluster_3	Developed ex-US Equity	Developed ex-US E...	61	FLAT				-0.513	-0.614
5	cluster_5	U.S. Large Cap Equity	U.S. Large Cap Eq...	55	FLAT	shifted			-0.761	-0.383
23	cluster_23	Bank Loan / Floating Rate	Bank Loan / Float...	9	FLAT				-0.942	-0.109
8	cluster_8	Asia-Pacific Equity	Asia-Pacific Equi...	25	FLAT				-0.978	-0.087
17	cluster_17	Gold	Gold	12	FLAT				-0.986	-0.079
13	cluster_13	China Equity	China Equity	15	FLAT				-0.990	-0.017
21	cluster_21	Inflation-Protected Fixed I...	Inflation-Protect...	10	FLAT	shifted			-0.991	-1.968
10	cluster_10	Materials	Materials	21	FLAT				-0.997	-0.030
19	cluster_19	Broad Commodities	Broad Commodities	11	LONG	shifted	14	-14.5%	-1.000	-1.035
25	cluster_25	International Fixed Income	International Fix...	6	LONG		17	-2.6%	-1.000	-0.003

Cluster Details

Cluster 1 – U.S. Mid Cap Equity

Members: 83 **Var explained:** 0.868 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +40.5%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +27.1% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-15 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Industrials	15	0.1687
U.S. Small Cap Equity	10	0.1266
U.S. Small Cap Value	6	0.0755
Consumer Discretionary	6	0.0667
U.S. Mid Cap Equity	5	0.0637
U.S. Mid Cap Value	3	0.0382
U.S. Small Cap Growth	3	0.0379
U.S. Mid Cap Growth	3	0.0372

Utilities	3	0.0337
Dividend Equity	2	0.0243
Leveraged / Inverse	2	0.0243
U.S. Small Cap Low Volatility	2	0.0232
U.S. Small Cap Fundamental Equity	1	0.0128
Unknown	1	0.0128
U.S. Small Cap ESG Equity	1	0.0128
U.S. Mid/Small Cap Equity	1	0.0128
Mid Cap Value Equities	1	0.0128
U.S. Small/Mid Cap Fundamental Equity	1	0.0127
U.S. Small Cap Multifactor Equity	1	0.0127
U.S. Mid Cap Dividend Equity	1	0.0125
Small Cap Value Equities	1	0.0125
U.S. Small/Mid Cap Equity	1	0.0125
U.S. ESG Equity	1	0.0124
U.S. Mid Cap Quality Equity	1	0.0123
U.S. Small Cap Dividend Equity	1	0.0123
Value Equity	1	0.0123
U.S. Equity	1	0.0123
U.S. Small Cap Industrials	1	0.0122
U.S. Small Cap Momentum Equity	1	0.0121
U.S. Micro Cap Equity	1	0.0117
U.S. Mid Cap Momentum Equity	1	0.0116
U.S. Dividend Equity	1	0.0116
U.S. Small Cap Consumer Discretionary	1	0.0115
U.S. Mid Cap Low Volatility	1	0.0112
Real Estate	1	0.0099

Replication basket – IJH (cor=0.99) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
IJH	0.994	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
16	FXN	0.0128	First Trust Mid Cap Core AlphaDEX Fund	U.S. Mid/Small Cap Equity	FLAT
19	NUSC	0.0128	Nuveen ESG Small-Cap ETF	U.S. Small Cap ESG Equity	FLAT
22	MDY	0.0128	State Street SPDR S&P MIDCAP 400 ETF Tr...	U.S. Mid Cap Equity	FLAT
29	IVOO	0.0128	Vanguard S&P Mid-Cap 400 ETF		FLAT
30	VI00	0.0128	Vanguard S&P Small-Cap 600 ETF	U.S. Small Cap Equity	FLAT

32	IJR	0.0128	iShares Core S&P Small-Cap ETF	U.S. Small Cap Equity	FLAT
33	IJH	0.0128	iShares Core S&P Mid-Cap ETF	U.S. Mid Cap Equity	FLAT
34	SPMD	0.0128	State Street SPDR Portfolio S&P 400 Mid...	U.S. Mid Cap Equity	FLAT
35	SPSM	0.0128	State Street SPDR Portfolio S&P 600 Sma...	U.S. Small Cap Equity	FLAT
36	VB	0.0128	Vanguard Small-Cap ETF	U.S. Small Cap Equity	FLAT
50	EZM	0.0128	WisdomTree U.S. MidCap Fund	Mid Cap Value Equities	FLAT
55	FNDA	0.0128	Schwab Fundamental U.S. Small Company E...	U.S. Small Cap Fundament...	FLAT
67	VBR	0.0128	Vanguard Small-Cap Value ETF	U.S. Small Cap Value	FLAT
15	SCHM	0.0127	Schwab U.S. Mid Cap ETF	U.S. Mid Cap Equity	LONG
17	FYX	0.0127	First Trust Small Cap Core AlphaDEX Fund	U.S. Small Cap Equity	FLAT
20	PRFZ	0.0127	Invesco RAFI US 1500 Small-Mid ETF	U.S. Small/Mid Cap Funda...	FLAT
21	SMLF	0.0127	iShares U.S. Small-Cap Equity Factor ETF	U.S. Small Cap Multifact...	FLAT
23	IJT	0.0127	iShares S&P SmallCap 600 Growth ETF	U.S. Small Cap Growth	FLAT
28	SLYG	0.0127	State Street SPDR S&P 600 Small Cap Gro...	U.S. Small Cap Growth	FLAT
37	SCHA	0.0127	Schwab U.S. Small-Cap ETF	U.S. Small Cap Equity	LONG
60	IJJ	0.0127	iShares S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
62	IVOV	0.0127	Vanguard S&P Mid-Cap 400 Value ETF	U.S. Mid Cap Value	FLAT
63	MDYV	0.0127	State Street SPDR S&P 400 Mid Cap Value...	U.S. Mid Cap Value	FLAT
5	IWM	0.0126	iShares Russell 2000 Index Fund	U.S. Small Cap Equity	FLAT
25	VIOG	0.0126	Vanguard S&P Small-Cap 600 Growth ETF	U.S. Small Cap Growth	FLAT
31	VTWO	0.0126	Vanguard Russell 2000 ETF	U.S. Small Cap Equity	FLAT
58	IWN	0.0126	iShares Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
66	VTWV	0.0126	Vanguard Russell 2000 Value ETF	U.S. Small Cap Value	FLAT
75	RWK	0.0126	Invesco S&P MidCap 400 Revenue ETF	U.S. Mid Cap Equity	FLAT
82	SDVY	0.0126	First Trust SMID Cap Rising Dividend Ac...	Dividend Equity	LONG
7	FXR	0.0125	First Trust Industrials AlphaDEX	Industrials	LONG
48	DON	0.0125	WisdomTree U.S. MidCap Dividend Fund	U.S. Mid Cap Dividend Eq...	FLAT
49	EES	0.0125	WisdomTree U.S. SmallCap Fund	Small Cap Value Equities	FLAT
51	PSC	0.0125	Principal U.S. Small-Cap ETF	U.S. Small Cap Equity	FLAT
61	IJS	0.0125	iShares S&P SmallCap 600 Value ETF	U.S. Small Cap Value	LONG
64	SLYV	0.0125	State Street SPDR S&P 600 Small Cap Val...	U.S. Small Cap Value	FLAT
65	VIOV	0.0125	Vanguard S&P Small-Cap 600 Value ETF	U.S. Small Cap Value	FLAT
83	ISMD	0.0125	Inspire Small/Mid Cap ETF	U.S. Small/Mid Cap Equity	LONG
24	IJK	0.0124	iShares S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
26	IVOG	0.0124	Vanguard S&P Mid-Cap 400 Growth ETF	U.S. Mid Cap Growth	LONG
27	MDYG	0.0124	State Street SPDR S&P 400 Mid Cap Growt...	U.S. Mid Cap Growth	LONG
68	ETHO	0.0124	Amplify Etho Climate Leadership U.S. ETF	U.S. ESG Equity	FLAT
69	RWJ	0.0124	Invesco S&P SmallCap 600 Revenue ETF	U.S. Small Cap Equity	FLAT
47	DES	0.0123	WisdomTree U.S. SmallCap Dividend Fund	U.S. Small Cap Dividend ...	FLAT
72	XMHQ	0.0123	Invesco S&P MidCap Quality ETF	U.S. Mid Cap Quality Equ...	FLAT
74	RFV	0.0123	Invesco S&P MidCap 400 Pure Value ETF	Leveraged / Inverse	FLAT
76	OMFS	0.0123	Invesco Russell 2000 Dynamic Multifacto...	U.S. Equity	FLAT
81	FYT	0.0123	First Trust Small Cap Value AlphaDEX Fu...	Value Equity	FLAT
70	PSCI	0.0122	Invesco S&P SmallCap Industrials ETF	U.S. Small Cap Industria...	FLAT
11	IYJ	0.0121	iShares U.S. Industrials ETF	Industrials	FLAT

13	VIS	0.0121	Vanguard Industrials ETF	Industrials	FLAT
14	FIDU	0.0121	Fidelity MSCI Industrials Index ETF	Industrials	FLAT
71	PAVE	0.0121	Global X U.S. Infrastructure Developmen...	Industrials	FLAT
80	XSMO	0.0121	Invesco S&P SmallCap Momentum ETF	U.S. Small Cap Momentum ...	FLAT
38	FXD	0.0120	First Trust Cons. Discret. AlphaDEX	Consumer Discretionary	LONG
77	RZV	0.0120	Invesco S&P SmallCap 600 Pure Value ETF	Leveraged / Inverse	FLAT
45	RSPD	0.0118	Invesco S&P 500 Equal Weight Consumer D...	Consumer Discretionary	LONG
59	SMMV	0.0118	iShares MSCI USA Small-Cap Min Vol Fact...	U.S. Small Cap Low Volat...	LONG
1	XLI	0.0117	State Street Industrial Select Sector S...	Industrials	FLAT
10	FIW	0.0117	First Trust Water ETF	Utilities	LONG
12	REGL	0.0117	ProShares S&P MidCap 400 Dividend Arist...	Dividend Equity	LONG
18	IWC	0.0117	iShares Microcap ETF	U.S. Micro Cap Equity	FLAT
6	XMMO	0.0116	Invesco S&P MidCap Momentum ETF	U.S. Mid Cap Momentum Eq...	LONG
40	FTXR	0.0116	First Trust Nasdaq Transportation ETF	Industrials	LONG
46	SMDV	0.0116	ProShares Russell 2000 Dividend Growers...	U.S. Dividend Equity	FLAT
79	PSCD	0.0115	Invesco S&P SmallCap Consumer Discretio...	U.S. Small Cap Consumer ...	LONG
8	PHO	0.0114	Invesco Water Resources ETF	Utilities	FLAT
54	XTN	0.0114	State Street SPDR S&P Transportation ETF	Industrials	FLAT
57	XSLV	0.0114	Invesco S&P SmallCap Low Volatility ETF	U.S. Small Cap Low Volat...	FLAT
73	PKB	0.0114	Invesco Building & Construction ETF	Industrials	LONG
56	XMLV	0.0112	Invesco S&P MidCap Low Volatility ETF	U.S. Mid Cap Low Volatil...	LONG
2	XRT	0.0111	State Street SPDR S&P Retail ETF	Consumer Discretionary	LONG
44	IYT	0.0111	iShares U.S. Transportation ETF	Industrials	FLAT
39	PEJ	0.0109	Invesco Leisure and Entertainment ETF	Consumer Discretionary	LONG
53	XHB	0.0109	State Street SPDR S&P Homebuilders ETF	Consumer Discretionary	LONG
78	PRN	0.0108	Invesco Dorsey Wright Industrials Momen...	Industrials	FLAT
9	CGW	0.0106	Invesco S&P Global Water Index ETF	Utilities	FLAT
52	XAR	0.0102	State Street SPDR S&P Aerospace & Defen...	Industrials	LONG
41	PPA	0.0101	Invesco Aerospace & Defense ETF	Industrials	LONG
3	JETS	0.0100	U.S. Global Jets ETF	Industrials	FLAT
43	ITB	0.0100	iShares U.S. Home Construction ETF	Consumer Discretionary	LONG
4	REM	0.0099	iShares Mortgage Real Estate ETF	Real Estate	FLAT
42	ITA	0.0095	iShares U.S. Aerospace & Defense ETF	Industrials	LONG

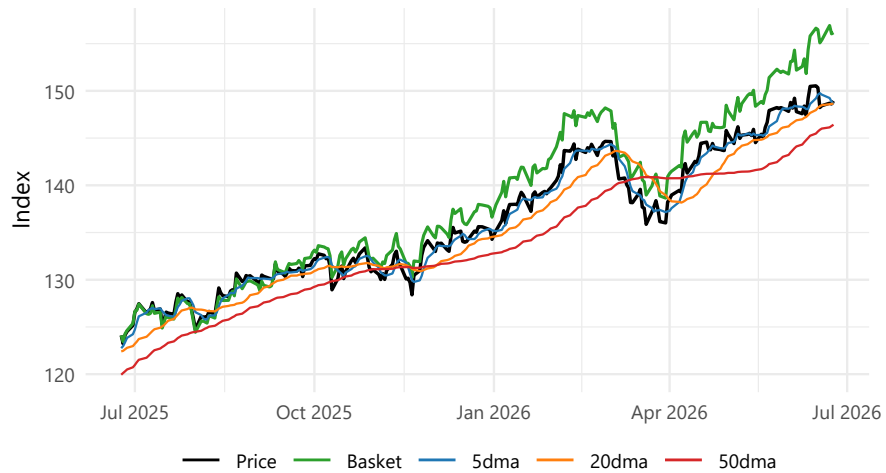
Cluster 2 – U.S. Large Cap Value

Members: 69 **Var explained:** 0.891 **Signal:** LONG since 2026-04-13 (50 days, 4.7%)

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +23.2%)
- Latest sell_cross: 2026-03-06

- Current 252d return: +20.9% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-04-13 > last sell at 2026-03-06)



Category breakdown

category	n_members	abs_weight
U.S. Dividend Equity	21	0.3016
U.S. Large Cap Value	10	0.1512
Value Equity	4	0.0584
U.S. Mid Cap Equity	4	0.0581
U.S. Low Volatility Equity	4	0.0539
U.S. Multifactor Equity	3	0.0443
U.S. Value Factor Equity	3	0.0434
U.S. Fundamental Equity	2	0.0302
U.S. Mid Cap Value	2	0.0300
U.S. Equal Weight Equity	2	0.0299
U.S. Equity	2	0.0298
Dividend Equity	2	0.0277
U.S. Mega Cap Value	1	0.0150
U.S. Revenue Weighted Equity	1	0.0149
Large Cap Blend Equities	1	0.0147
U.S. Size Factor Equity	1	0.0147
U.S. Buyback Equity	1	0.0145
U.S. Large Cap Equity	1	0.0144
U.S. Quality / Factor Equity	1	0.0141

U.S. Defensive Equity	1	0.0138
Global Equity	1	0.0133
Communication Services	1	0.0122

Replication basket – VTV (cor=0.98) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
VTV	1.006	Vanguard Value ETF	U.S. Large Cap Value	LONG

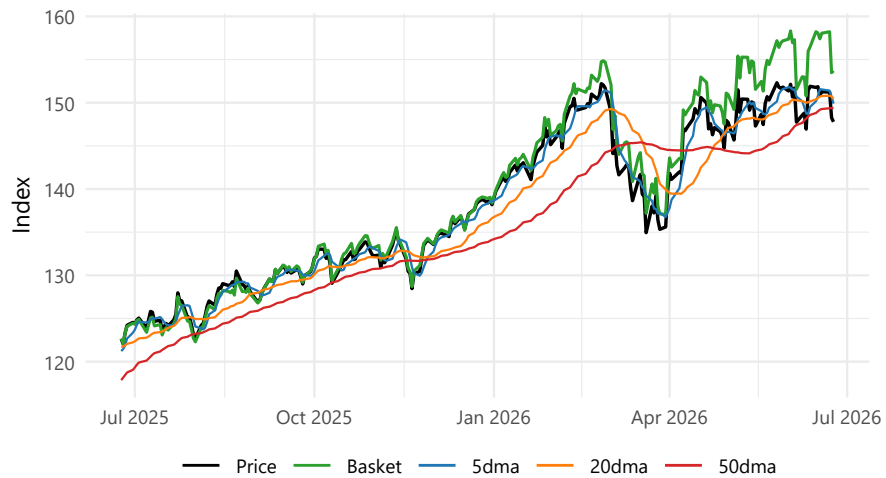
Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
25	IWD	0.0152	iShares Russell 1000 Value ETF	U.S. Large Cap Value	LONG
28	JPUS	0.0152	JPMorgan Diversified Return U.S. Equity...	U.S. Multifactor Equity	FLAT
33	VONV	0.0152	Vanguard Russell 1000 Value ETF	U.S. Large Cap Value	LONG
40	SCHV	0.0152	Schwab U.S. Large-Cap Value ETF	U.S. Large Cap Value	LONG
2	RSP	0.0151	Invesco S&P 500 Equal Weight ETF	U.S. Equal Weight Equity	LONG
10	PRF	0.0151	Invesco RAFI US 1000 ETF	U.S. Fundamental Equity	LONG
17	NULV	0.0151	Nuveen ESG Large-Cap Value ETF	U.S. Large Cap Value	FLAT
21	FNDX	0.0151	Schwab Fundamental U.S. Large Company E...	U.S. Fundamental Equity	LONG
27	IVE	0.0151	iShares S&P 500 Value ETF	U.S. Large Cap Value	FLAT
31	VOOV	0.0151	Vanguard S&P 500 Value ETF	U.S. Large Cap Value	LONG
37	ILCV	0.0151	iShares Morningstar Large-Cap Value ETF	U.S. Large Cap Value	FLAT
39	IUSV	0.0151	iShares Core S&P U.S. Value ETF	U.S. Large Cap Value	FLAT
41	SPYV	0.0151	State Street SPDR Portfolio S&P 500 Val...	U.S. Large Cap Value	FLAT
43	VTV	0.0151	Vanguard Value ETF	U.S. Large Cap Value	LONG
55	VOE	0.0151	Vanguard Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
60	DTD	0.0151	WisdomTree U.S. Total Dividend Fund	U.S. Dividend Equity	LONG
65	DGRO	0.0151	iShares Core Dividend Growth ETF	U.S. Dividend Equity	LONG
15	CFA	0.0150	VictoryShares US 500 Volatility Wtd ETF	U.S. Multifactor Equity	FLAT
23	IWX	0.0150	iShares Russell Top 200 Value ETF	Value Equity	LONG
34	MGV	0.0150	Vanguard Mega Cap Value ETF	U.S. Mega Cap Value	LONG
36	VYM	0.0150	Vanguard High Dividend Yield ETF	U.S. Dividend Equity	FLAT
59	DLN	0.0150	WisdomTree U.S. LargeCap Dividend Fund	U.S. Dividend Equity	LONG
11	RWL	0.0149	Invesco S&P 500 Revenue ETF	U.S. Revenue Weighted Eq...	LONG
13	AIVL	0.0149	WisdomTree U.S. AI Enhanced Value Fund	U.S. Value Factor Equity	LONG
24	ONEV	0.0149	State Street SPDR Russell 1000 Low Vol...	U.S. Equity	FLAT
44	FTA	0.0149	First Trust Large Cap Value AlphaDEX Fu...	Value Equity	LONG
52	IWS	0.0149	iShares Russell Mid-Cap Value ETF	U.S. Mid Cap Value	FLAT
57	PFM	0.0149	Invesco Dividend Achievers ETF	U.S. Dividend Equity	FLAT
68	JPME	0.0149	JPMorgan Diversified Return U.S. Mid Ca...	U.S. Equity	FLAT

22	EQAL	0.0148	Invesco Russell 1000 Equal Weight ETF	U.S. Equal Weight Equity	FLAT
53	ONEY	0.0148	State Street SPDR Russell 1000 Yield Fo...	U.S. Dividend Equity	FLAT
5	FEX	0.0147	First Trust Large Cap Core AlphaDEX Fund	Large Cap Blend Equities	LONG
19	FDVV	0.0147	Fidelity High Dividend ETF	U.S. Dividend Equity	FLAT
29	SIZE	0.0147	iShares MSCI USA Size Factor ETF	U.S. Size Factor Equity	LONG
66	VIG	0.0147	Vanguard Div Appreciation ETF	U.S. Dividend Equity	LONG
9	JHMM	0.0146	John Hancock Multifactor Mid Cap ETF	U.S. Mid Cap Equity	FLAT
18	SDY	0.0146	State Street SPDR S&P Dividend ETF	U.S. Dividend Equity	LONG
35	SCHD	0.0146	Schwab US Dividend Equity ETF	U.S. Dividend Equity	FLAT
38	IMCB	0.0146	iShares Morningstar Mid-Cap ETF	U.S. Mid Cap Equity	LONG
4	PKW	0.0145	Invesco BuyBack Achievers ETF	U.S. Buyback Equity	LONG
6	PWV	0.0145	Invesco Large Cap Value ETF	Value Equity	LONG
26	IWR	0.0145	iShares Russell Mid-Cap ETF	U.S. Mid Cap Equity	LONG
42	VO	0.0145	Vanguard Mid-Cap ETF	U.S. Mid Cap Equity	LONG
47	SDOG	0.0145	ALPS Sector Dividend Dogs ETF	U.S. Dividend Equity	FLAT
48	DVY	0.0145	iShares Select Dividend ETF	U.S. Dividend Equity	FLAT
69	JVAL	0.0145	JPMorgan U.S. Value Factor ETF	U.S. Value Factor Equity	LONG
3	FVD	0.0144	First Trust VL Dividend	U.S. Dividend Equity	LONG
62	DIA	0.0144	State Street SPDR Dow Jones Industrial ...	U.S. Large Cap Equity	LONG
7	RDVY	0.0143	First Trust Rising Dividend Achievers E...	U.S. Dividend Equity	FLAT
54	SPYD	0.0142	State Street SPDR Portfolio S&P 500 Hig...	U.S. Dividend Equity	FLAT
58	OUSA	0.0142	ALPS O'Shares U.S. Quality Dividend ETF	Dividend Equity	FLAT
64	LGLV	0.0142	State Street SPDR US Large Cap Low Vola...	U.S. Low Volatility Equi...	LONG
16	CFO	0.0141	VictoryShares US 500 Enhanced Volatilit...	U.S. Multifactor Equity	FLAT
50	RPV	0.0141	Invesco S&P 500 Pure Value ETF	Value Equity	FLAT
56	FTCS	0.0141	First Trust Capital Strength ETF	U.S. Quality / Factor Eq...	LONG
12	DHS	0.0140	WisdomTree U.S. High Dividend Fund	U.S. Dividend Equity	FLAT
30	VLUE	0.0140	iShares MSCI USA Value Factor ETF	U.S. Value Factor Equity	LONG
14	CDC	0.0138	VictoryShares US EQ Income Enhanced Vol...	U.S. Defensive Equity	FLAT
63	USMV	0.0138	iShares MSCI USA Min Vol Factor ETF	U.S. Low Volatility Equi...	FLAT
49	RDIV	0.0137	Invesco S&P Ultra Dividend Revenue ETF	U.S. Dividend Equity	FLAT
51	SPHD	0.0136	Invesco S&P 500 High Dividend Low Volat...	Dividend Equity	LONG
8	FDL	0.0135	First Trust Morningstar ETF	U.S. Dividend Equity	FLAT
45	PEY	0.0135	Invesco High Yield Equity Dividend Achi...	U.S. Dividend Equity	FLAT
1	ACWV	0.0133	iShares MSCI Global Min Vol Factor ETF	Global Equity	FLAT
46	DIV	0.0133	Global X Super Dividend ETF	U.S. Dividend Equity	FLAT
20	LVHD	0.0132	Franklin U.S. Low Volatility High Divid...	U.S. Low Volatility Equi...	LONG
32	HDV	0.0132	iShares Core High Dividend ETF	U.S. Dividend Equity	FLAT
61	SPLV	0.0127	Invesco S&P 500 Low Volatility ETF	U.S. Low Volatility Equi...	LONG
67	IYZ	0.0122	iShares U.S. Telecommunications ETF	Communication Services	FLAT

Cluster 3 – Developed ex-US Equity**Members:** 61 **Var explained:** 0.896 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +47.7%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +22.0% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-10)

**Category breakdown**

category	n_members	abs_weight
International Equity	19	0.3158
Developed ex-US Equity	9	0.1528
Global ex-US Equity	5	0.0848
Europe Equity	4	0.0675
International Dividend Equity	4	0.0659
Alternatives	4	0.0615
Eurozone Equity	3	0.0490
Commodity	1	0.0172
Global ex-US Small Cap Equity	1	0.0166
Germany Equity	1	0.0162
France Equity	1	0.0162

Italy Equity	1	0.0159
United Kingdom Equity	1	0.0158
Europe Financials	1	0.0158
Netherlands Equity	1	0.0157
Sweden Equity	1	0.0156
Spain Equity	1	0.0150
Austria Equity	1	0.0148
Belgium Equity	1	0.0146
Poland Equity	1	0.0135

Replication basket – VEA (cor=0.99) basket lead at lag 1: -0.016

ticker	rep_wt	name	category	etf_signal
VEA	0.963	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	EFA	0.0172	iShares MSCI EAFE ETF	Developed ex-US Equity	FLAT
3	VEA	0.0172	Vanguard FTSE Developed Markets ETF	Developed ex-US Equity	FLAT
40	GSIE	0.0172	Goldman Sachs ActiveBeta International ...	Commodity	LONG
41	ESGD	0.0172	iShares ESG Aware MSCI EAFE ETF	International Equity	FLAT
51	IEFA	0.0172	iShares Core MSCI EAFE ETF	Developed ex-US Equity	FLAT
52	SCHF	0.0172	Schwab International Equity ETF	Developed ex-US Equity	FLAT
53	SPDW	0.0172	State Street SPDR Portfolio Developed W...	Developed ex-US Equity	FLAT
25	TLTD	0.0171	FlexShares Morningstar Developed Market...	International Equity	FLAT
6	IEV	0.0170	iShares Europe ETF	Europe Equity	FLAT
16	DWM	0.0170	WisdomTree International Equity Fund	International Dividend E...	LONG
19	PXF	0.0170	Invesco RAFI Developed Markets ex-U.S. ...	International Equity	FLAT
29	ACWX	0.0170	iShares MSCI ACWI ex U.S. ETF	Global ex-US Equity	FLAT
32	INTF	0.0170	iShares International Equity Factor ETF	International Equity	FLAT
46	IEUR	0.0170	iShares Core MSCI Europe ETF	Europe Equity	FLAT
47	IXUS	0.0170	iShares Core MSCI Total International S...	Global ex-US Equity	FLAT
48	VEU	0.0170	Vanguard FTSE All World Ex US ETF	Global ex-US Equity	FLAT
49	VGK	0.0170	Vanguard FTSEEuropean ETF	Europe Equity	FLAT
50	VXUS	0.0170	Vanguard Total International Stock ETF	Global ex-US Equity	FLAT
30	CWI	0.0169	State Street SPDR MSCI ACWI ex-US ETF	Global ex-US Equity	FLAT
33	IQLT	0.0169	iShares MSCI Intl Quality Factor ETF	International Equity	LONG
35	QEFA	0.0169	State Street SPDR MSCI EAFE StrategicFa...	International Equity	FLAT
39	FNDF	0.0169	Schwab Fundamental International Equity...	International Equity	FLAT
14	EZU	0.0168	iShares MSCI Eurozone ETF	Eurozone Equity	FLAT

23	EFV	0.0168	iShares MSCI EAFE Value ETF	Developed ex-US Equity	FLAT
18	IQDF	0.0167	FlexShares International Quality Divide...	International Equity	FLAT
22	SCZ	0.0167	iShares MSCI EAFE Small-Cap ETF	Developed ex-US Equity	FLAT
24	FNDC	0.0167	Schwab Fundamental International Small ...	International Equity	FLAT
26	JPIN	0.0167	JPMorgan Diversified Return Internation...	International Equity	FLAT
37	RODM	0.0167	Hartford Multifactor Developed Markets ...	International Equity	FLAT
38	VYMI	0.0167	Vanguard International High Dividend Yi...	International Dividend E...	FLAT
44	SCHC	0.0167	Schwab International Small-Cap Equity E...	Developed ex-US Equity	FLAT
56	HFXI	0.0167	NYLI FTSE International Equity Currency...	International Equity	FLAT
20	EFG	0.0166	iShares MSCI EAFE Growth ETF	Developed ex-US Equity	FLAT
34	IVLU	0.0166	iShares MSCI Intl Value Factor ETF	International Equity	FLAT
36	FEZ	0.0166	State Street SPDR EURO STOXX 50 ETF	Eurozone Equity	FLAT
45	VSS	0.0166	Vanguard FTSE All-Wld ex-US SmCp Idx ETF	Global ex-US Small Cap E...	FLAT
4	FEP	0.0165	First Trust Europe AlphaDEX Fund	Europe Equity	FLAT
7	DLS	0.0165	WisdomTree International SmallCap Fund	International Equity	FLAT
31	IMTM	0.0164	iShares MSCI Intl Momentum Factor ETF	International Equity	FLAT
43	VIGI	0.0163	Vanguard International Dividend Appreci...	International Dividend E...	FLAT
1	EWG	0.0162	iShares MSCI Germany Index Fund	Germany Equity	FLAT
12	EWQ	0.0162	iShares MSCI France Index Fund	France Equity	FLAT
21	GWX	0.0162	State Street SPDR S&P International Sma...	International Equity	FLAT
5	VIDI	0.0160	Vident International Equity Strategy ETF	International Equity	FLAT
59	IPKW	0.0160	Invesco International BuyBack Achievers...	International Equity	FLAT
61	EWI	0.0159	iShares MSCI Italy ETF	Italy Equity	LONG
13	EWU	0.0158	iShares MSCI United Kingdom ETF	United Kingdom Equity	FLAT
15	IDV	0.0158	iShares International Select Dividend E...	International Dividend E...	FLAT
17	EUFN	0.0158	iShares MSCI Europe Financials ETF	Europe Financials	LONG
27	DBEF	0.0157	Xtrackers MSCI EAFE Hedged Equity ETF	Alternatives	FLAT
42	HDEF	0.0157	Xtrackers MSCI EAFE High Dividend Yield...	International Equity	FLAT
58	EWN	0.0157	iShares MSCI Netherlands Index Fund	Netherlands Equity	LONG
10	HEZU	0.0156	iShares Currency Hedged MSCI Eurozone E...	Eurozone Equity	FLAT
28	HEFA	0.0156	iShares Currency Hedged MSCI EAFE ETF	Alternatives	FLAT
55	EWD	0.0156	iShares MSCI Sweden ETF	Sweden Equity	FLAT
8	HEDJ	0.0151	WisdomTree Europe Hedged Equity Fund	Alternatives	FLAT
9	IHDG	0.0151	WisdomTree International Hedged Quality...	Alternatives	LONG
11	EWP	0.0150	iShares MSCI Spain ETF	Spain Equity	FLAT
57	EWO	0.0148	iShares MSCI Austria ETF	Austria Equity	LONG
60	EWK	0.0146	iShares MSCI Belgium ETF	Belgium Equity	FLAT
54	EPOL	0.0135	iShares MSCI Poland ETF	Poland Equity	FLAT

Cluster 4 – Aggregate Fixed Income

Members: 58 Var explained: 0.832 Signal: LONG since 2026-06-17 (4 days, 0.5%)

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +4.5%)
- Latest sell_cross: 2026-05-13
- Current 252d return: +4.3% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-17 > last sell at 2026-05-13)

**Category breakdown**

category	n_members	abs_weight
Fixed Income	19	0.3283
Investment Grade Corporate Bond	13	0.2302
Aggregate Fixed Income	6	0.1120
Mortgage-Backed Fixed Income	6	0.0982
Active Fixed Income	5	0.0891
Cash / Ultra-Short Fixed Income	3	0.0470
International Fixed Income	2	0.0313
Target Maturity Bond	2	0.0307
Agency Fixed Income	1	0.0167
Taxable Municipal Bond	1	0.0165

Replication basket – BND (cor=0.99) basket lead at lag 1: 0.006

ticker	rep_wt	name	category	etf_signal
BND	1.007	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
21	BIV	0.0188	Vanguard Intermediate-Term Bond ETF	Fixed Income	FLAT
26	AGG	0.0188	iShares Core U.S. Aggregate Bond ETF	Aggregate Fixed Income	LONG
28	BND	0.0188	Vanguard Total Bond Market ETF	Aggregate Fixed Income	LONG
30	SCHZ	0.0188	Schwab US Aggregate Bond ETF	Aggregate Fixed Income	LONG
20	IUSB	0.0187	iShares Core Universal USD Bond ETF	Aggregate Fixed Income	LONG
31	SPAB	0.0187	State Street SPDR Portfolio Aggregate B...	Aggregate Fixed Income	LONG
2	IEF	0.0185	iShares 7-10 Year Treasury Bond ETF	Fixed Income	FLAT
17	IGIB	0.0184	iShares 5-10 Year Investment Grade Corp...	Investment Grade Corpora...	FLAT
24	VCIT	0.0184	Vanguard Intermediate-Term Corporate Bo...	Investment Grade Corpora...	FLAT
7	FBND	0.0183	Fidelity Total Bond ETF	Active Fixed Income	LONG
43	SPTI	0.0183	State Street SPDR Portfolio Intermediat...	Fixed Income	FLAT
46	SCHR	0.0183	Schwab Intermediate-Term U.S. Treasury ...	Fixed Income	FLAT
47	VGIT	0.0183	Vanguard Intermediate-Term Treasury ETF	Fixed Income	FLAT
13	AGGY	0.0182	WisdomTree Yield Enhanced U.S. Aggregat...	Aggregate Fixed Income	LONG
15	SPIB	0.0182	State Street SPDR Portfolio Intermediat...	Investment Grade Corpora...	FLAT
6	GTO	0.0181	Invesco Total Return Bond ETF	Active Fixed Income	LONG
11	GVI	0.0181	iShares Intermediate Government/Credit ...	Fixed Income	FLAT
40	IEI	0.0181	iShares 3-7 Year Treasury Bond ETF	Fixed Income	FLAT
50	MBB	0.0181	iShares MBS ETF	Mortgage-Backed Fixed In...	LONG
12	QLTA	0.0180	iShares Aaa A Rated Corporate Bond ETF	Investment Grade Corpora...	LONG
27	USIG	0.0180	iShares Broad USD Investment Grade Corp...	Investment Grade Corpora...	LONG
51	VMBS	0.0180	Vanguard Mortgage-Backed Securities ETF	Mortgage-Backed Fixed In...	LONG
52	SPMB	0.0180	State Street SPDR Portfolio Mortgage Ba...	Mortgage-Backed Fixed In...	LONG
3	LQD	0.0179	iShares iBoxx \$ Investment Grade Corpor...	Investment Grade Corpora...	LONG
5	BOND	0.0179	PIMCO Active Bond Exchange-Traded Fund ...	Active Fixed Income	LONG
19	ISTB	0.0178	iShares Core 1-5 Year USD Bond ETF	Fixed Income	FLAT
8	FLCO	0.0177	Franklin Investment Grade Corporate ETF	Active Fixed Income	LONG
10	CORP	0.0177	PIMCO Investment Grade Corporate Bond I...	Investment Grade Corpora...	LONG
22	BLV	0.0176	Vanguard Long-Term Bond ETF	Fixed Income	LONG
41	TLH	0.0176	iShares 10-20 Year Treasury Bond ETF	Fixed Income	LONG
45	GOVT	0.0176	iShares U.S. Treasury Bond ETF	Fixed Income	FLAT
35	IGSB	0.0175	iShares 1-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
37	VCSH	0.0175	Vanguard Short-Term Corporate Bond ETF	Investment Grade Corpora...	FLAT
23	BSV	0.0174	Vanguard Short-Term Bond ETF	Fixed Income	FLAT
49	GNMA	0.0174	iShares GNMA Bond ETF	Mortgage-Backed Fixed In...	FLAT
18	IGLB	0.0172	iShares 10 Year Investment Grade Corpor...	Investment Grade Corpora...	LONG

16	SPLB	0.0171	State Street SPDR Portfolio Long Term C...	Investment Grade Corpora...	LONG
25	VCLT	0.0171	Vanguard Long-Term Corporate Bond ETF	Investment Grade Corpora...	LONG
36	SLQD	0.0171	iShares 0-5 Year Investment Grade Corpo...	Investment Grade Corpora...	FLAT
44	SPTL	0.0171	State Street SPDR Portfolio Long Term T...	Fixed Income	LONG
48	VGLT	0.0171	Vanguard Long-Term Treasury ETF	Fixed Income	LONG
29	TOTL	0.0170	State Street DoubleLine Total Return Ta...	Active Fixed Income	FLAT
1	TLT	0.0169	iShares 20+ Year Treasury Bond ETF	Fixed Income	LONG
39	AGZ	0.0167	iShares Agency Bond ETF	Agency Fixed Income	FLAT
9	BAB	0.0165	Invesco Taxable Municipal Bond ETF	Taxable Municipal Bond	LONG
42	EDV	0.0163	Vanguard Extended Duration Treasury ETF	Fixed Income	LONG
34	SPSB	0.0162	State Street SPDR Portfolio Short Term ...	Fixed Income	FLAT
53	SHY	0.0162	iShares 1-3 Year Treasury Bond ETF	Fixed Income	FLAT
56	VGSH	0.0159	Vanguard Short-Term Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
57	BNDX	0.0158	Vanguard Total International Bond ETF	International Fixed Inco...	LONG
33	BSCQ	0.0156	Invesco BulletShares 2026 Corporate Bon...	Target Maturity Bond	FLAT
55	SCHO	0.0156	Schwab Short-Term U.S. Treasury ETF	Cash / Ultra-Short Fixed...	FLAT
58	IAGG	0.0156	iShares International Aggregate Bond Fu...	International Fixed Inco...	LONG
54	SPTS	0.0155	State Street SPDR Portfolio Short Term ...	Cash / Ultra-Short Fixed...	FLAT
14	IBDR	0.0151	iShares iBonds Dec 2026 Term Corporate ...	Target Maturity Bond	FLAT
38	CMBS	0.0136	iShares CMBS Bond ETF	Mortgage-Backed Fixed In...	FLAT
4	LMBS	0.0133	First Trust Low Duration Opportunities ...	Mortgage-Backed Fixed In...	LONG
32	LDUR	0.0121	PIMCO Enhanced Low Duration Active Exch...	Fixed Income	LONG

Cluster 5 – U.S. Large Cap Equity

Members: 55 **Var explained:** 0.923 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +24.9%)
- Latest sell_cross: 2026-06-24
- Current 252d return: +19.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-24 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
U.S. Large Cap Equity	9	0.1697
U.S. Total Market Equity	7	0.1321
Global Equity	7	0.1293
Consumer Discretionary	5	0.0849
Asset Allocation	4	0.0685
U.S. Mega Cap Equity	3	0.0555
U.S. Quality / Factor Equity	3	0.0549
U.S. Multifactor Equity	3	0.0548
Communication Services	3	0.0493
U.S. ESG Equity	2	0.0376
U.S. Equity	2	0.0376
Commodity	1	0.0188
Large Cap Value Equities	1	0.0187
Large Cap Blend Equities	1	0.0186
Dividend Equity	1	0.0183
U.S. Dividend Equity	1	0.0180
U.S. Moat Equity	1	0.0172
Alternatives	1	0.0160

Replication basket – VOO (cor=1.00) basket lead at lag 1: -0.011

ticker	rep_wt	name	category	etf_signal
V00	0.96	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
28	IWV	0.0189	iShares Russell 3000 Fund	U.S. Total Market Equity	FLAT
29	IYY	0.0189	iShares Dow Jones U.S. ETF	U.S. Total Market Equity	FLAT
34	ESGU	0.0189	iShares ESG Aware MSCI USA ETF	U.S. ESG Equity	FLAT
35	IWB	0.0189	iShares Russell 1000 ETF	U.S. Large Cap Equity	FLAT
43	VONE	0.0189	Vanguard Russell 1000 ETF	U.S. Large Cap Equity	FLAT
45	VV	0.0189	Vanguard Large-Cap ETF	U.S. Large Cap Equity	FLAT
46	ILCB	0.0189	iShares Morningstar Large-Cap ETF	U.S. Large Cap Equity	FLAT
47	ITOT	0.0189	iShares Core S&P Total U.S. Stock Marke...	U.S. Total Market Equity	FLAT
48	IVV	0.0189	iShares Core S&P 500 ETF	U.S. Large Cap Equity	FLAT
49	SCHB	0.0189	Schwab U.S. Broad Market ETF	U.S. Total Market Equity	FLAT
50	SCHX	0.0189	Schwab U.S. Large-Cap ETF	U.S. Large Cap Equity	FLAT
51	SPTM	0.0189	State Street SPDR Portfolio S&P 1500 Co...	U.S. Total Market Equity	FLAT
52	V00	0.0189	Vanguard S&P 500 ETF	U.S. Large Cap Equity	FLAT
53	VTI	0.0189	Vanguard Total Stock Market ETF	U.S. Total Market Equity	FLAT
3	SPY	0.0188	State Street SPDR S&P 500 ETF Trust	U.S. Large Cap Equity	FLAT
32	SPYX	0.0188	State Street SPDR S&P 500 Fossil Fuel R...	U.S. Equity	FLAT
40	VTHR	0.0188	Vanguard Russell 3000 ETF	U.S. Total Market Equity	FLAT
41	GSLC	0.0188	Goldman Sachs ActiveBeta U.S. Large Cap...	Commodity	FLAT
10	URTH	0.0187	iShares MSCI World ETF	Global Equity	FLAT
24	JHML	0.0187	John Hancock Multifactor Large Cap ETF	U.S. Large Cap Equity	FLAT
27	SUSA	0.0187	iShares ESG Optimized MSCI USA ETF	U.S. ESG Equity	FLAT
30	LRGF	0.0187	iShares U.S. Equity Factor ETF	U.S. Multifactor Equity	FLAT
36	IWL	0.0187	iShares Russell Top 200 ETF	U.S. Equity	FLAT
42	EPS	0.0187	WisdomTree U.S. LargeCap Fund	Large Cap Value Equities	FLAT
44	MGC	0.0187	Vanguard Mega Cap ETF	U.S. Mega Cap Equity	FLAT
5	ACWI	0.0186	iShares MSCI ACWI ETF	Global Equity	FLAT
13	VT	0.0186	Vanguard Total World Stock Index ETF	Global Equity	FLAT
20	TILT	0.0186	FlexShares Mornigstar US Market Factors...	U.S. Multifactor Equity	FLAT
26	DSI	0.0186	iShares ESG MSCI KLD 400 ETF	Large Cap Blend Equities	FLAT
31	OEF	0.0186	iShares S&P 100 Fund	U.S. Mega Cap Equity	FLAT
11	CRBN	0.0185	iShares Low Carbon Optimized MSCI ACWI ...	Global Equity	FLAT
12	SPGM	0.0185	State Street SPDR Portfolio MSCI Global...	Global Equity	FLAT
37	QUAL	0.0185	iShares MSCI USA Quality Factor ETF	U.S. Quality / Factor Eq...	LONG
6	AOA	0.0184	iShares Core 80/20 Aggressive Allocatio...	Asset Allocation	FLAT
22	QDF	0.0183	FlexShares Quality Dividend Index Fund	Dividend Equity	FLAT
38	QUS	0.0183	State Street SPDR MSCI USA StrategicFac...	U.S. Quality / Factor Eq...	FLAT

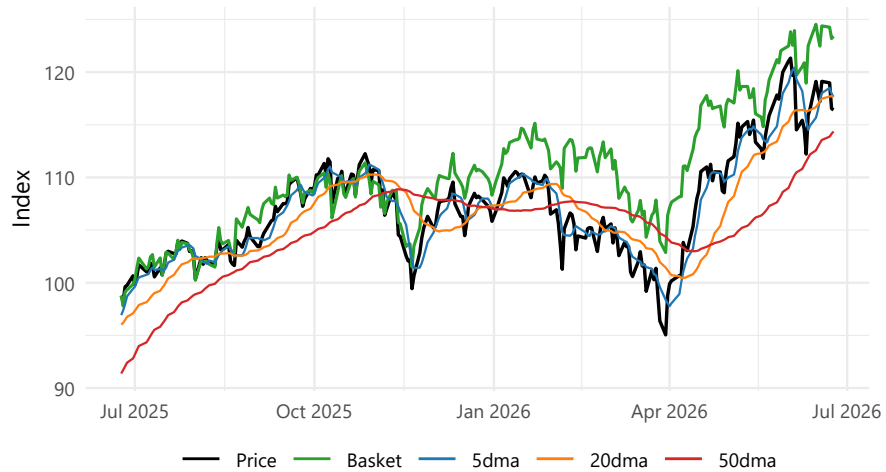
21	I00	0.0182	iShares Global 100 ETF	Global Equity	FLAT
23	CATH	0.0182	Global X S&P 500 Catholic Values ETF	Global Equity	FLAT
33	XLG	0.0182	Invesco S&P 500 Top 50 ETF	U.S. Mega Cap Equity	FLAT
25	DGRW	0.0180	WisdomTree U.S. Quality Dividend Growth...	U.S. Dividend Equity	FLAT
39	SPHQ	0.0180	Invesco S&P 500 Quality ETF	U.S. Quality / Factor Eq...	LONG
9	AOR	0.0179	iShares Core 60/40 Balanced Allocation ...	Asset Allocation	FLAT
14	IYC	0.0176	iShares U.S. Consumer Discretionary ETF	Consumer Discretionary	FLAT
55	OMFL	0.0175	Invesco Russell 1000 Dynamic Multifacto...	U.S. Multifactor Equity	FLAT
15	VCR	0.0172	Vanguard Consumer Discretion ETF	Consumer Discretionary	FLAT
18	FDIS	0.0172	Fidelity MSCI Consumer Discretionary In...	Consumer Discretionary	FLAT
19	MOAT	0.0172	VanEck Morningstar Wide Moat ETF	U.S. Moat Equity	FLAT
1	XLY	0.0170	State Street Consumer Discretionary Sel...	Consumer Discretionary	FLAT
8	AOM	0.0168	iShares Core 40/60 Moderate Allocation ...	Asset Allocation	FLAT
16	VOX	0.0166	Vanguard Communication Services ETF	Communication Services	FLAT
17	FCOM	0.0166	Fidelity MSCI Communication Services In...	Communication Services	FLAT
2	XLC	0.0162	State Street Communication Services Sel...	Communication Services	FLAT
4	QAI	0.0160	NYLI Hedge Multi-Strategy Tracker ETF	Alternatives	FLAT
54	RTH	0.0159	VanEck Retail ETF	Consumer Discretionary	FLAT
7	AOK	0.0154	iShares Core 30/70 Conservative Allocat...	Asset Allocation	FLAT

Cluster 6 – U.S. Extended Market Equity

Members: 41 **Var explained:** 0.807 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +26.7%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +19.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Technology	16	0.3816
U.S. Mid Cap Growth	3	0.0793
U.S. Small Cap Growth	3	0.0783
U.S. Momentum Equity	3	0.0705
Growth Equity	2	0.0514
IPO Equity	2	0.0504
Convertible Bond	2	0.0500
Financials	2	0.0486
Momentum Equity	2	0.0484
U.S. Extended Market Equity	1	0.0264
U.S. Equity	1	0.0256
Consumer Discretionary	1	0.0241
U.S. Small Cap Momentum Equity	1	0.0238
International IPO Equity	1	0.0217
Israel Equity	1	0.0200

Replication basket – VXF (cor=0.97) basket lead at lag 1: 0.017

ticker	rep_wt	name	category	etf_signal
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VXF 1.013 Vanguard Extended Market ETF U.S. Extended Market ... FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
26	IWP	0.0265	iShares Russell Midcap Growth ETF	U.S. Mid Cap Growth	LONG
34	VBK	0.0265	Vanguard Small-Cap Growth ETF	U.S. Small Cap Growth	FLAT
29	VOT	0.0264	Vanguard Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
30	IMCG	0.0264	iShares Morningstar Mid-Cap Growth ETF	U.S. Mid Cap Growth	LONG
36	VXF	0.0264	Vanguard Extended Market ETF	U.S. Extended Market Equ...	FLAT
35	ISCG	0.0261	iShares Morningstar Small-Cap Growth ETF	Growth Equity	FLAT
32	IWO	0.0259	iShares Russell 2000 Growth Fund	U.S. Small Cap Growth	LONG
33	VTWG	0.0259	Vanguard Russell 2000 Growth ETF	U.S. Small Cap Growth	LONG
12	XT	0.0258	iShares Future Exponential Technologies...	Technology	FLAT
23	FPX	0.0257	First Trust US Equity Opportunities ETF	IPO Equity	FLAT
37	SPHB	0.0256	Invesco S&P 500 High Beta ETF	U.S. Equity	LONG
21	FTC	0.0254	First Trust Large Cap Growth AlphaDEX F...	Technology	LONG
24	CWB	0.0253	State Street SPDR Bloomberg Convertible...	Convertible Bond	FLAT
25	RPG	0.0253	Invesco S&P 500 Pure Growth ETF	Growth Equity	LONG
4	ROBO	0.0251	ROBO Global Robotics and Automation Ind...	Technology	FLAT
22	PDP	0.0250	Invesco Dorsey Wright Momentum ETF	Momentum Equity	LONG
20	FINX	0.0249	Global X FinTech ETF	Financials	FLAT
9	ARKQ	0.0248	ARK Autonomous Technology & Robotics ETF	Technology	LONG
27	ICVT	0.0248	iShares Convertible Bond ETF	Convertible Bond	FLAT
3	BOTZ	0.0247	Global X Robotics & Artificial Intellig...	Technology	FLAT
15	IPO	0.0247	Renaissance IPO ETF	IPO Equity	LONG
39	VFMO	0.0247	Vanguard U.S. Momentum Factor ETF	U.S. Momentum Equity	LONG
17	FDN	0.0245	First Trust DJ Internet Index Fund	Technology	FLAT
2	SKYY	0.0243	First Trust Cloud Computing ETF	Technology	FLAT
8	ARKW	0.0243	ARK Next Generation Internet ETF	Technology	FLAT
7	ARKK	0.0242	ARK Innovation ETF	Technology	LONG
16	PNQI	0.0242	Invesco Nasdaq Internet ETF	Technology	FLAT
13	IBUY	0.0241	Amplify Online Retail ETF	Consumer Discretionary	LONG
31	DWAS	0.0238	Invesco Dorsey Wright SmallCap Momentum...	U.S. Small Cap Momentum ...	FLAT
18	IPAY	0.0237	Amplify Digital Payments ETF	Financials	FLAT
11	PRNT	0.0236	3D Printing (The) ETF	Technology	FLAT
1	IGV	0.0234	iShares Expanded Tech-Software Sector E...	Technology	FLAT
10	FV	0.0234	First Trust Dorsey Wright Focus 5 ETF	Momentum Equity	LONG
28	MTUM	0.0233	iShares MSCI USA Momentum Factor ETF	U.S. Momentum Equity	LONG
14	CIBR	0.0229	First Trust NASDAQ Cybersecurity ETF	Technology	FLAT
6	HACK	0.0227	Amplify Cybersecurity ETF	Technology	FLAT
38	QMOM	0.0224	Alpha Architect U.S. Quantitative Momen...	U.S. Momentum Equity	FLAT
5	BLOK	0.0223	Amplify Blockchain Technology ETF	Technology	FLAT

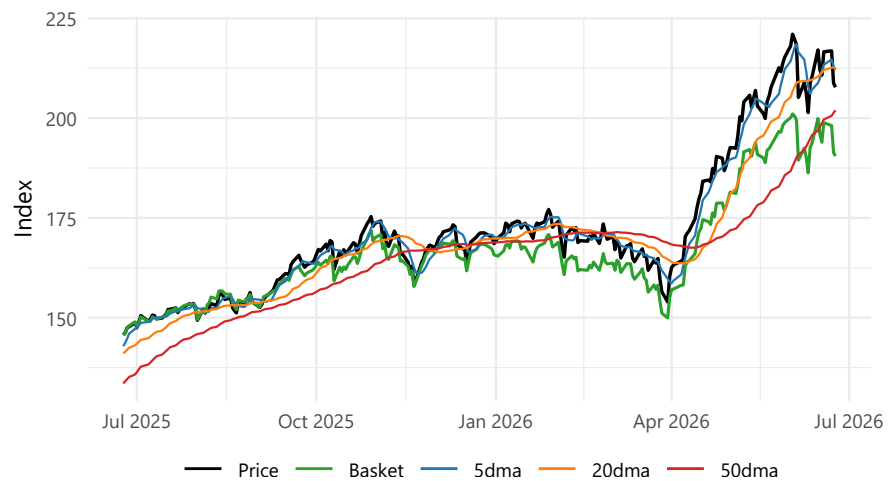
19	FPXI	0.0217	First Trust International Equity Opport...	International IPO Equity	FLAT
40	EIS	0.0200	iShares MSCI Israel ETF	Israel Equity	FLAT
41	BLCN	0.0195	Siren NexGen Economy ETF	Technology	LONG

Cluster 7 – U.S. Large Cap Growth

Members: 39 **Var explained:** 0.896 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +55.3%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +45.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Technology	19	0.4862
U.S. Large Cap Growth	11	0.2915
Growth Equity	2	0.0522

U.S. Equity	1	0.0267
U.S. Mega Cap Growth	1	0.0264
U.S. Equal Weight Equity	1	0.0257
Options Income	1	0.0239
U.S. Small Cap Technology	1	0.0236
Trend-Following Equity	1	0.0235
U.S. Trend-Following Equity	1	0.0204

Replication basket – QQQ (cor=0.98) basket lead at lag 1: -0.030

ticker	rep_wt	name	category	etf_signal
QQQ	1.023	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	QQQ	0.0268	Invesco QQQ Trust, Series 1	U.S. Large Cap Growth	FLAT
17	IGM	0.0268	iShares Expanded Tech Sector ETF	Technology	FLAT
25	VGT	0.0268	Vanguard Information Tech ETF	Technology	FLAT
27	FTEC	0.0268	Fidelity MSCI Information Technology In...	Technology	FLAT
19	IYW	0.0267	iShares U.S. Technology ETF	Technology	FLAT
21	ONEQ	0.0267	Fidelity Nasdaq Composite Index ETF	U.S. Equity	FLAT
1	XLK	0.0266	State Street Technology Select Sector S...	Technology	FLAT
18	IXN	0.0266	iShares Global Tech ETF	Technology	FLAT
30	ILCG	0.0266	iShares Morningstar Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
23	IWF	0.0265	iShares Russell 1000 Growth Fund	U.S. Large Cap Growth	FLAT
24	IVW	0.0265	iShares S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
26	VOOG	0.0265	Vanguard S&P 500 Growth ETF	U.S. Large Cap Growth	FLAT
28	VONG	0.0265	Vanguard Russell 1000 Growth ETF	U.S. Large Cap Growth	FLAT
31	IUSG	0.0265	iShares Core S&P U.S. Growth ETF	U.S. Large Cap Growth	FLAT
33	SPYG	0.0265	State Street SPDR Portfolio S&P 500 Gro...	U.S. Large Cap Growth	FLAT
34	VUG	0.0265	Vanguard Growth ETF	U.S. Large Cap Growth	FLAT
13	XNTK	0.0264	State Street SPDR NYSE Technology ETF	Technology	LONG
20	NULG	0.0264	Nuveen ESG Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
29	MGK	0.0264	Vanguard Mega Cap Growth ETF	U.S. Mega Cap Growth	FLAT
32	SCHG	0.0264	Schwab U.S. Large-Cap Growth ETF	U.S. Large Cap Growth	FLAT
22	IWY	0.0263	iShares Russell Top 200 Growth ETF	Growth Equity	FLAT
8	QTEC	0.0262	First Trust NASDAQ-100-Technology Secto...	Technology	LONG
12	FXL	0.0260	First Trust Technology AlphaDEX	Technology	LONG
4	AIQ	0.0259	Global X Artificial Intelligence & Tech...	Technology	FLAT
16	PWB	0.0259	Invesco Large Cap Growth ETF	Growth Equity	FLAT

7	QQEW	0.0257	First Trust Nasdaq-100 Select Equal Wei...	U.S. Equal Weight Equity	LONG
10	TDIV	0.0253	First Trust NASDAQ Technology Dividend ...	Technology	FLAT
3	SMH	0.0252	VanEck Semiconductor ETF	Technology	LONG
2	SOXX	0.0249	iShares PHLX SOX Semiconductor Sector I...	Technology	LONG
39	QTUM	0.0248	Defiance Quantum ETF	Technology	LONG
9	PSI	0.0244	Invesco Semiconductors ETF	Technology	LONG
11	XSD	0.0243	State Street SPDR S&P Semiconductor ETF	Technology	LONG
37	FTXL	0.0243	First Trust Nasdaq Semiconductor ETF	Technology	LONG
6	NXTG	0.0241	First Trust Indxx NextG ETF	Technology	LONG
38	PTF	0.0241	Invesco Dorsey Wright Technology Moment...	Technology	LONG
15	QYLD	0.0239	Global X NASDAQ 100 Covered Call ETF	Options Income	FLAT
36	PSCT	0.0236	Invesco S&P SmallCap Information Techno...	U.S. Small Cap Technology	LONG
14	PTNQ	0.0235	Pacer Trendpilot 100 ETF	Trend-Following Equity	FLAT
35	PTLC	0.0204	Pacer Trendpilot US Large Cap ETF	U.S. Trend-Following Equ...	FLAT

Cluster 8 – Asia-Pacific Equity

Members: 25 **Var explained:** 0.672 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-22 – filter **PASS** (252d at date = +16.3%)
- Latest sell_cross: 2026-06-22
- Current 252d return: +11.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-22 more recent than last effective buy 2026-05-22)



Category breakdown

category	n_members	abs_weight
Asia-Pacific Equity	2	0.0914
International Equity	2	0.0887
International Dividend Equity	2	0.0887
Real Estate	2	0.0851
Global Equity	2	0.0849
Emerging Markets Fixed Income	2	0.0774
Currency	2	0.0750
Australia Equity	1	0.0452
Canada Equity	1	0.0430
Switzerland Equity	1	0.0400
Singapore Equity	1	0.0389
South Africa Equity	1	0.0382
New Zealand Equity	1	0.0366
Latin America Equity	1	0.0366
Mexico Equity	1	0.0346
Consumer Discretionary	1	0.0339
Brazil Equity	1	0.0308
Chile Equity	1	0.0308

Replication basket – EWZ+IGF+BBAX (cor=0.97) basket lead at lag 1: -0.005

ticker	rep_wt	name	category	etf_signal
EWZ	0.088	iShares MSCI Brazil ETF	Brazil Equity	FLAT
IGF	0.272	iShares Global Infrastructure ETF	Global Equity	FLAT
BBAX	0.482	JPMorgan BetaBuilders Developed Asia ...	Asia-Pacific Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

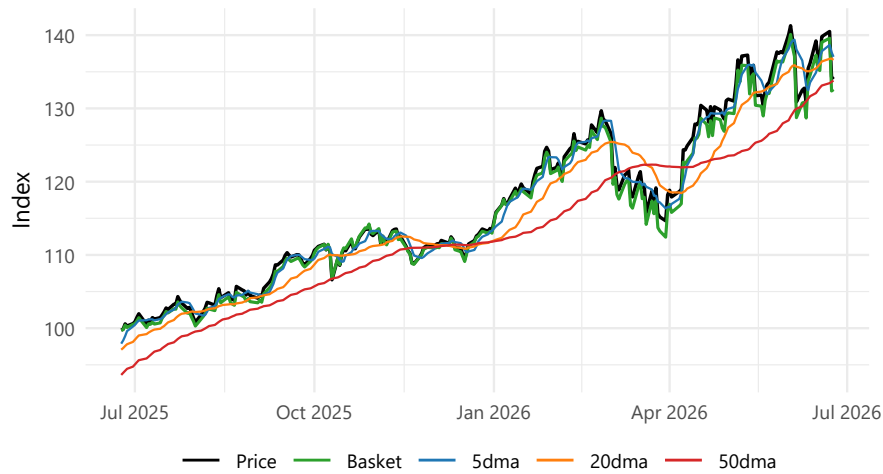
	ticker	PC_wt	name	category	etf_signal
9	EPP	0.0458	iShares MSCI Pacific Ex-Japan Index Fund	Asia-Pacific Equity	FLAT
25	BBAX	0.0457	JPMorgan BetaBuilders Developed Asia Pa...	Asia-Pacific Equity	FLAT
13	IDLV	0.0453	Invesco S&P International Developed Low...	International Equity	FLAT
5	EWA	0.0452	iShares MSCI Australia Index Fund	Australia Equity	FLAT
12	DWX	0.0446	State Street SPDR S&P International Div...	International Dividend E...	FLAT
4	PID	0.0441	Invesco International Dividend Achiever...	International Dividend E...	FLAT
14	EFAV	0.0435	iShares MSCI EAFE Min Vol Factor ETF	International Equity	FLAT
15	VNQI	0.0432	Vanguard Global ex-U.S. Real Estate ETF	Real Estate	FLAT
10	NFRA	0.0431	FlexShares STOXX Global Broad Infrastru...	Global Equity	FLAT
8	EWC	0.0430	iShares MSCI Canada Index Fund	Canada Equity	FLAT
3	RWX	0.0420	State Street SPDR Dow Jones Internation...	Real Estate	FLAT
11	IGF	0.0419	iShares Global Infrastructure ETF	Global Equity	FLAT
6	EWL	0.0400	iShares MSCI Switzerland ETF	Switzerland Equity	FLAT
21	FXA	0.0399	Invesco CurrencyShares Australian Dolla...	Currency	FLAT
7	EWS	0.0389	iShares MSCI Singapore ETF	Singapore Equity	LONG
16	EBND	0.0388	State Street SPDR Bloomberg Emerging Ma...	Emerging Markets Fixed I...	LONG
17	EMLC	0.0386	VanEck J. P. Morgan EM Local Currency B...	Emerging Markets Fixed I...	LONG
23	EZA	0.0382	iShares MSCI South Africa Index Fund	South Africa Equity	FLAT
20	ILF	0.0366	iShares Latin America 40 ETF	Latin America Equity	FLAT
22	ENZL	0.0366	iShares MSCI New Zealand ETF	New Zealand Equity	FLAT
24	FXC	0.0351	Invesco CurrencyShares Canadian Dollar ...	Currency	FLAT
18	EWX	0.0346	iShares MSCI Mexico ETF	Mexico Equity	FLAT
1	ESPO	0.0339	VanEck Video Gaming and eSports ETF	Consumer Discretionary	FLAT
2	EWZ	0.0308	iShares MSCI Brazil ETF	Brazil Equity	FLAT
19	ECH	0.0308	iShares MSCI Chile ETF	Chile Equity	FLAT

Cluster 9 – Emerging Markets Equity

Members: 25 **Var explained:** 0.873 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +57.0%)
- Latest sell_cross: 2026-06-10
- Current 252d return: +37.4% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-10 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Emerging Markets Equity	19	0.7726
Asia Equity	2	0.0813
Asia ex-Japan Equity	1	0.0418
Taiwan Equity	1	0.0361
South Korea Equity	1	0.0341
Global Dividend Equity	1	0.0340

Replication basket – IEMG (cor=0.99) basket lead at lag 1: -0.028

ticker	rep_wt	name	category	etf_signal
IEMG	0.923	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

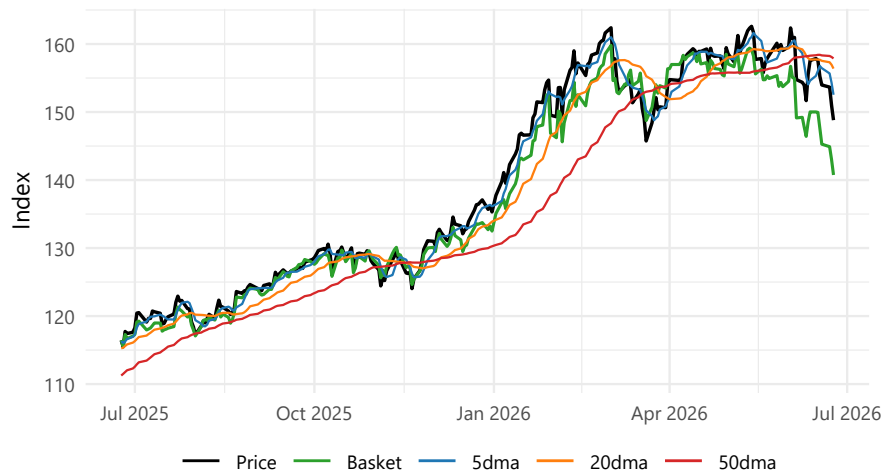
	ticker	PC_wt	name	category	etf_signal
21	IEMG	0.0425	iShares Core MSCI Emerging Markets ETF	Emerging Markets Equity	FLAT
2	EEM	0.0424	iShares MSCI Emerging Index Fund	Emerging Markets Equity	FLAT
3	VWO	0.0422	Vanguard FTSE Emerging Markets ETF	Emerging Markets Equity	LONG
16	GEM	0.0422	Goldman Sachs ActiveBeta Emerging Marke...	Emerging Markets Equity	FLAT
20	ESGE	0.0422	iShares ESG Aware MSCI EM ETF	Emerging Markets Equity	FLAT
23	SPEM	0.0422	State Street SPDR Portfolio Emerging Ma...	Emerging Markets Equity	FLAT
22	SCHE	0.0421	Schwab Emerging Markets Equity ETF	Emerging Markets Equity	LONG
18	XSOE	0.0420	WisdomTree Emerging Markets Ex-State Ow...	Emerging Markets Equity	FLAT
5	AAXJ	0.0418	iShares MSCI All Country Asia ex Japan ...	Asia ex-Japan Equity	FLAT
15	EMGF	0.0418	iShares Emerging Markets Equity Factor ...	Emerging Markets Equity	FLAT
11	EEMA	0.0414	iShares MSCI Emerging Markets Asia ETF	Emerging Markets Equity	FLAT
14	GMF	0.0411	State Street SPDR S&P Emerging Asia Pac...	Asia Equity	FLAT
17	FNDE	0.0410	Schwab Fundamental Emerging Markets Equ...	Emerging Markets Equity	FLAT
12	PXH	0.0409	Invesco RAFI Emerging Markets ETF	Emerging Markets Equity	FLAT
19	EEMV	0.0405	iShares MSCI Emerging Markets Min Vol F...	Emerging Markets Equity	FLAT
8	DGS	0.0402	WisdomTree Emerging Market SmallCap Fund	Emerging Markets Equity	LONG
10	AIA	0.0402	iShares Asia 50 ETF	Asia Equity	FLAT
7	DEM	0.0396	WisdomTree Emerging Markets High Divide...	Emerging Markets Equity	LONG
6	EWX	0.0392	State Street SPDR S&P Emerging Markets ...	Emerging Markets Equity	LONG
4	FEM	0.0388	First Trust Emerging Markets AlphaDEX F...	Emerging Markets Equity	FLAT
13	DVYE	0.0365	iShares Emerging Markets Dividend Index...	Emerging Markets Equity	FLAT
9	EWI	0.0361	iShares MSCI Taiwan ETF	Taiwan Equity	LONG
25	FEMS	0.0348	First Trust Emerging Markets Small Cap ...	Emerging Markets Equity	FLAT
1	EWY	0.0341	iShares MSCI South Korea ETF	South Korea Equity	FLAT
24	SDIV	0.0340	Global X SuperDividend ETF	Global Dividend Equity	FLAT

Cluster 10 – Materials

Members: 21 **Var explained:** 0.703 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +44.4%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +28.2% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-09 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Materials	13	0.6543
MLP / Energy Infrastructure	4	0.1687
Energy	1	0.0501
Consumer Staples	1	0.0493
Norway Equity	1	0.0440
Industrial Metals	1	0.0336

Replication basket – AMLP+URA+GUNR (cor=0.96) basket lead at lag 1: 0.044

ticker	rep_wt	name	category	etf_signal
AMLP	0.109	Alerian MLP ETF	MLP / Energy Infrastr...	FLAT
URA	0.083	Global X Uranium ETF	Materials	FLAT
GUNR	0.775	FlexShares Global Upstream Natural Re...	Materials	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
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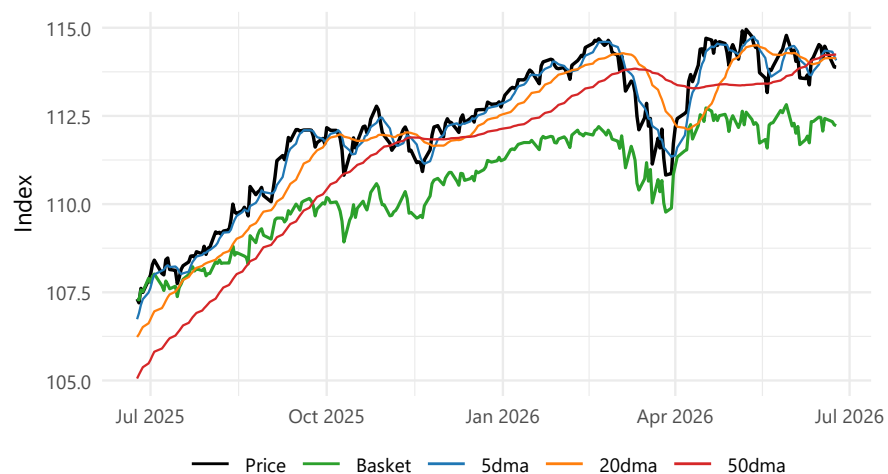
15	GNR	0.0550	State Street SPDR S&P Global Natural Re...	Materials	FLAT
14	GUNR	0.0543	FlexShares Global Upstream Natural Reso...	Materials	FLAT
10	MXI	0.0529	iShares Global Materials ETF	Materials	FLAT
11	IYM	0.0529	iShares U.S. Basic Materials ETF	Materials	LONG
21	NANR	0.0525	State Street SPDR S&P North American Na...	Materials	FLAT
13	VAW	0.0513	Vanguard Materials ETF	Materials	LONG
16	PICK	0.0509	iShares MSCI Global Select Metals & Min...	Materials	FLAT
12	RSPM	0.0505	Invesco S&P 500 Equal Weight Materials ...	Materials	LONG
1	XLB	0.0504	State Street Materials Select Sector SP...	Materials	LONG
20	IGE	0.0501	iShares North American Natural Resource...	Energy	FLAT
4	XME	0.0499	State Street SPDR S&P Metals & Mining E...	Materials	FLAT
17	SLX	0.0498	VanEck Steel ETF	Materials	FLAT
6	MOO	0.0493	VanEck Agribusiness ETF	Consumer Staples	FLAT
3	COPX	0.0472	Global X Copper Miners ETF	Materials	FLAT
18	NORW	0.0440	Global X MSCI Norway ETF	Norway Equity	FLAT
5	EMLP	0.0437	First Trust North American Energy Infra...	MLP / Energy Infrastruct...	FLAT
9	MLPX	0.0436	Global X MLP & Energy Infrastructure ETF	MLP / Energy Infrastruct...	LONG
7	AML	0.0413	Alerian MLP ETF	MLP / Energy Infrastruct...	FLAT
8	MLPA	0.0401	Global X MLP ETF	MLP / Energy Infrastruct...	LONG
2	URA	0.0368	Global X Uranium ETF	Materials	FLAT
19	DBB	0.0336	Invesco DB Base Metals Fund	Industrial Metals	FLAT

Cluster 11 – High Yield Bond

Members: 21 **Var explained:** 0.766 **Signal:** LONG since 2026-06-17 (4 days, -0.2%)

Signal logic:

- Latest buy_cross: 2026-06-17 – filter **PASS** (252d at date = +7.3%)
- Latest sell_cross: 2026-06-09
- Current 252d return: +6.5% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-17 > last sell at 2026-06-09)



Category breakdown

category	n_members	abs_weight
High Yield Bond	10	0.5155
Preferred Stock	7	0.3061
Emerging Markets Fixed Income	4	0.1784

Replication basket – HYG (cor=0.95) basket lead at lag 1: 0.047

ticker	rep_wt	name	category	etf_signal
HYG	0.917	iShares iBoxx \$ High Yield Corporate ...	High Yield Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	HYG	0.0526	iShares iBoxx \$ High Yield Corporate Bo...	High Yield Bond	LONG
13	JNK	0.0526	State Street SPDR Bloomberg High Yield ...	High Yield Bond	LONG
19	HYLB	0.0524	Xtrackers USD High Yield Corporate Bond...	High Yield Bond	LONG
18	BBHY	0.0523	JPMorgan BetaBuilders USD High Yield Co...	High Yield Bond	LONG
16	SHYG	0.0521	iShares 0-5 Year High Yield Corporate B...	High Yield Bond	LONG

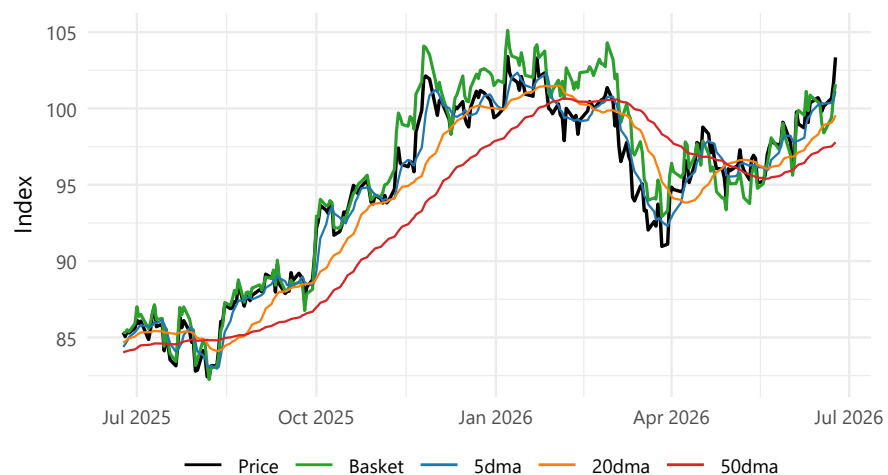
14	SJNK	0.0520	State Street SPDR Bloomberg Short Term ...	High Yield Bond	LONG
15	ANGL	0.0516	VanEck Fallen Angel High Yield Bond ETF	High Yield Bond	LONG
17	FALN	0.0516	iShares Fallen Angels USD Bond ETF	High Yield Bond	LONG
6	HYS	0.0511	PIMCO 0-5 Year High Yield Corporat Bond...	High Yield Bond	LONG
9	PFF	0.0479	iShares Preferred and Income Securities...	Preferred Stock	FLAT
3	HYLS	0.0473	First Trust Tactical High Yield ETF	High Yield Bond	LONG
2	EMB	0.0467	iShares J.P. Morgan USD Emerging Market...	Emerging Markets Fixed I...	FLAT
21	VWOB	0.0467	Vanguard Emerging Markets Government Bo...	Emerging Markets Fixed I...	LONG
11	PFXF	0.0461	VanEck Preferred Securities ex Financia...	Preferred Stock	FLAT
20	PCY	0.0457	Invesco Emerging Markets Sovereign Debt...	Emerging Markets Fixed I...	LONG
7	PGX	0.0446	Invesco Preferred ETF	Preferred Stock	FLAT
10	PSK	0.0441	State Street SPDR ICE Preferred Securit...	Preferred Stock	FLAT
4	FPE	0.0434	First Trust Preferred Securities and In...	Preferred Stock	LONG
5	PGF	0.0431	Invesco Financial Preferred ETF	Preferred Stock	FLAT
12	HYEM	0.0392	VanEck Emerging Markets High Yield Bond...	Emerging Markets Fixed I...	LONG
8	VRP	0.0368	Invesco Variable Rate Preferred ETF	Preferred Stock	FLAT

Cluster 12 – Healthcare

Members: 17 **Var explained:** 0.760 **Signal:** LONG since 2026-05-06 (33 days, 6.2%)

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +13.8%)
- Latest sell_cross: 2026-04-29
- Current 252d return: +23.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-06 > last sell at 2026-04-29)



Category breakdown

category	n_members	abs_weight
Healthcare	16	0.9418
U.S. Small Cap Healthcare	1	0.0582

Replication basket – VHT (cor=0.93) basket lead at lag 1: -0.004

ticker	rep_wt	name	category	etf_signal
VHT	1.099	Vanguard Health Care ETF	Healthcare	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
15	VHT	0.0654	Vanguard Health Care ETF	Healthcare	LONG
16	FHLC	0.0653	Fidelity MSCI Health Care Index ETF	Healthcare	LONG
9	FXH	0.0646	First Trust Health Care AlphaDEX	Healthcare	LONG
14	RSPH	0.0640	Invesco S&P 500 Equal Weight Health Car...	Healthcare	LONG
13	IYH	0.0638	iShares U.S. Healthcare ETF	Healthcare	LONG
10	IXJ	0.0623	iShares Global Healthcare ETF	Healthcare	FLAT

1	XLV	0.0621	State Street Health Care Select Sector ...	Healthcare	LONG
3	IBB	0.0604	iShares Biotechnology ETF	Healthcare	LONG
6	FBT	0.0596	First Trust Amex Biotech Index Fund	Healthcare	FLAT
8	PSCH	0.0582	Invesco S&P SmallCap Health Care ETF	U.S. Small Cap Healthcare	LONG
7	XHE	0.0574	State Street SPDR S&P Health Care Equip...	Healthcare	LONG
12	IHI	0.0570	iShares U.S. Medical Devices ETF	Healthcare	FLAT
17	PPH	0.0543	VanEck Pharmaceutical ETF	Healthcare	LONG
5	PTH	0.0542	Invesco Dorsey Wright Healthcare Moment...	Healthcare	LONG
4	XBI	0.0534	State Street SPDR S&P Biotech ETF	Healthcare	LONG
2	ARKG	0.0508	ARK Genomic Revolution ETF	Healthcare	FLAT
11	IHF	0.0472	iShares U.S. Health Care Providers ETF	Healthcare	LONG

Cluster 13 – China Equity

Members: 15 **Var explained:** 0.801 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-15 – filter **PASS** (252d at date = +24.3%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +6.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-04-15)



Category breakdown

category	n_members	abs_weight
China Equity	13	0.8766
Emerging Markets Equity	1	0.0643
Hong Kong Equity	1	0.0591

Replication basket – KWEB (cor=0.92) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
KWEB	0.563	KraneShares CSI China Internet ETF	China Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

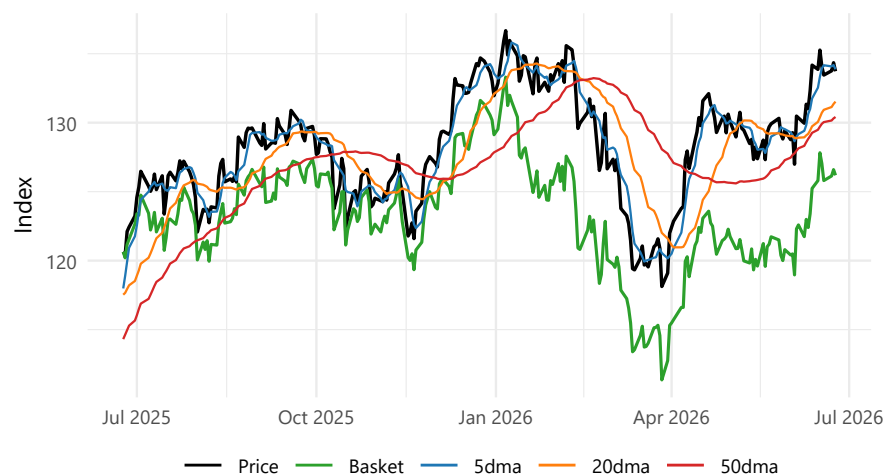
	ticker	PC_wt	name	category	etf_signal
11	CXSE	0.0737	WisdomTree China ex-State-Owned Enterpr...	China Equity	FLAT
9	GXC	0.0735	State Street SPDR S&P China ETF	China Equity	FLAT
1	MCHI	0.0730	iShares MSCI China ETF	China Equity	FLAT
2	FXI	0.0712	iShares China Large-Cap ETF	China Equity	FLAT
5	CQQQ	0.0711	Invesco China Technology ETF	China Equity	FLAT
7	CHIQ	0.0705	Global X MSCI China Consumer Discretion...	China Equity	FLAT
4	KWEB	0.0680	KraneShares CSI China Internet ETF	China Equity	FLAT
6	ASHR	0.0669	Xtrackers Harvest CSI 300 China A-Share...	China Equity	LONG
8	CNYA	0.0661	iShares MSCI China A ETF	China Equity	LONG
13	KGRN	0.0651	KraneShares MSCI China Clean Technology...	China Equity	FLAT
3	EMQQ	0.0643	EMQQ The Emerging Markets Internet ETF	Emerging Markets Equity	FLAT
10	KBA	0.0596	KraneShares Boseri MSCI China A 50 Conn...	China Equity	FLAT
14	CNXT	0.0594	VanEck ChiNext Innovators ETF	China Equity	FLAT
12	EWK	0.0591	iShares MSCI Hong Kong Index Fund	Hong Kong Equity	FLAT
15	KURE	0.0585	KraneShares MSCI All China Health Care ...	China Equity	FLAT

Cluster 14 – Financials

Members: 14 **Var explained:** 0.876 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-13 – filter **PASS** (252d at date = +19.9%)
- Latest sell_cross: 2026-05-11
- Current 252d return: +12.3% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-11 more recent than last effective buy 2026-04-13)



Category breakdown

category	n_members	abs_weight
Financials	13	0.9382
Global Equity	1	0.0618

Replication basket – XLF (cor=0.96) basket lead at lag 1: 0.011

ticker	rep_wt	name	category	etf_signal
XLF	1.097	State Street Financial Select Sector ...	Financials	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
10	VFH	0.0755	Vanguard Financials ETF	Financials	FLAT

11	FNCL	0.0755	Fidelity MSCI Financials Index ETF	Financials	LONG
5	FXO	0.0752	First Trust Financials AlphaDEX	Financials	LONG
7	IYF	0.0752	iShares U.S. Financial ETF	Financials	FLAT
8	IYG	0.0747	iShares U.S. Financial Services ETF	Financials	FLAT
1	XLF	0.0741	State Street Financial Select Sector SP...	Financials	LONG
9	KBWB	0.0733	Invesco KBW Bank ETF	Financials	FLAT
3	KBE	0.0712	State Street SPDR S&P Bank ETF	Financials	FLAT
14	KCE	0.0707	State Street SPDR S&P Capital Markets E...	Financials	FLAT
6	IAT	0.0706	iShares U.S. Regional Banks ETF	Financials	LONG
13	IAI	0.0698	iShares U.S. Broker-Dealers & Securitie...	Financials	LONG
2	KRE	0.0687	State Street SPDR S&P Regional Banking ...	Financials	FLAT
4	KIE	0.0638	State Street SPDR S&P Insurance ETF	Financials	LONG
12	PSP	0.0618	Invesco Global Listed Private Equity ETF	Global Equity	FLAT

Cluster 15 – Real Estate

Members: 14 **Var explained:** 0.961 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-10 – filter **PASS** (252d at date = +21.2%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +11.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-04-10)



Category breakdown

category	n_members	abs_weight
Real Estate	14	1

Replication basket – VNQ (cor=0.99) basket lead at lag 1: 0.000

ticker	rep_wt	name	category	etf_signal
VNQ	0.971	Vanguard Real Estate ETF	Real Estate	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
8	VNQ	0.0725	Vanguard Real Estate ETF	Real Estate	FLAT
9	FREL	0.0725	Fidelity MSCI Real Estate Index ETF	Real Estate	FLAT
11	SCHH	0.0725	Schwab U.S. REIT ETF	Real Estate	FLAT
2	IYR	0.0724	iShares U.S. Real Estate ETF	Real Estate	FLAT
10	USRT	0.0724	iShares Core U.S. REIT ETF	Real Estate	FLAT
6	RWR	0.0723	State Street SPDR Dow Jones REIT ETF	Real Estate	FLAT
13	PSR	0.0722	Invesco Active U.S. Real Estate Fund	Real Estate	FLAT

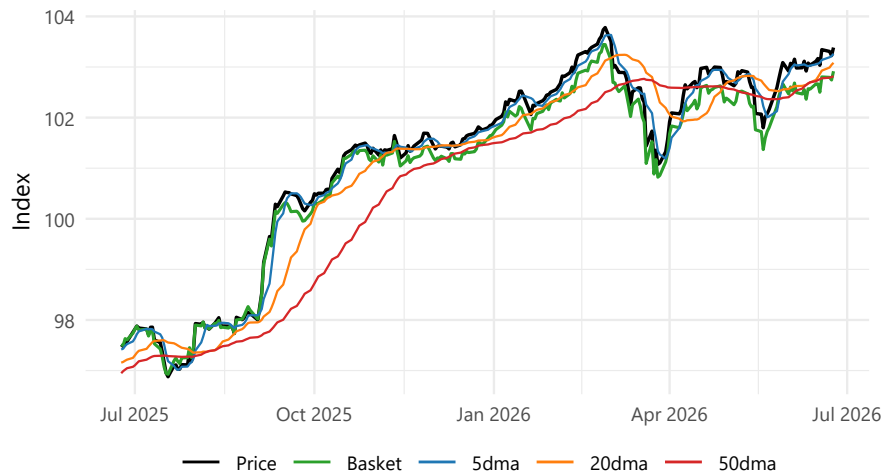
1	XLRE	0.0719	State Street Real Estate Select Sector ...	Real Estate	FLAT
5	ICF	0.0719	iShares Select U.S. REIT ETF	Real Estate	FLAT
7	REET	0.0715	iShares Global REIT ETF	Real Estate	FLAT
3	RWO	0.0711	State Street SPDR Dow Jones Global Real...	Real Estate	LONG
4	REZ	0.0698	iShares Residential and Multisector Rea...	Real Estate	FLAT
14	NURE	0.0688	Nuveen Short-Term REIT ETF	Real Estate	LONG
12	INDS	0.0682	Pacer Industrial Real Estate ETF	Real Estate	LONG

Cluster 16 – Municipal Bond

Members: 13 **Var explained:** 0.819 **Signal:** LONG since 2026-05-28 (18 days, 0.4%)

Signal logic:

- Latest buy_cross: 2026-05-28 – filter **PASS** (252d at date = +5.9%)
- Latest sell_cross: 2026-05-15
- Current 252d return: +6.0% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-05-28 > last sell at 2026-05-15)



Category breakdown

category	n_members	abs_weight
Municipal Bond	13	1

Replication basket – MUB (cor=0.95) basket lead at lag 1: 0.178

ticker	rep_wt	name	category	etf_signal
MUB	0.904	iShares National Muni Bond ETF	Municipal Bond	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

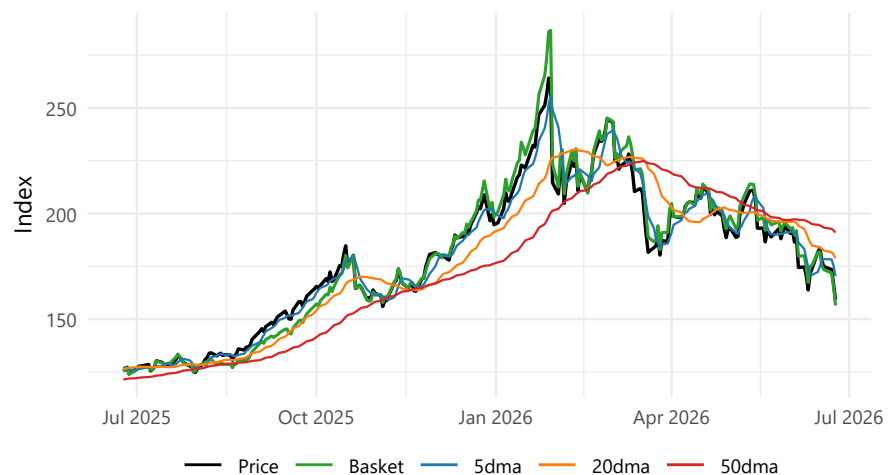
	ticker	PC_wt	name	category	etf_signal
13	VTEB	0.0817	Vanguard Tax-Exempt Bond ETF	Municipal Bond	LONG
11	MUB	0.0813	iShares National Muni Bond ETF	Municipal Bond	LONG
6	CMF	0.0802	iShares California Muni Bond ETF	Municipal Bond	LONG
9	TFI	0.0801	State Street SPDR Nuveen ICE Municipal ...	Municipal Bond	LONG
8	ITM	0.0800	VanEck Intermediate Muni ETF	Municipal Bond	LONG
1	FMB	0.0795	First Trust Managed Municipal ETF	Municipal Bond	LONG
7	NYF	0.0793	iShares New York Muni Bond ETF	Municipal Bond	LONG
5	PZA	0.0780	Invesco National AMT-Free Municipal Bon...	Municipal Bond	LONG
4	MUNI	0.0769	PIMCO Intermediate Municipal Bond Activ...	Municipal Bond	LONG
2	HYD	0.0748	VanEck High Yield Muni ETF	Municipal Bond	LONG
3	HYMB	0.0720	State Street SPDR Nuveen ICE High Yield...	Municipal Bond	LONG
12	SUB	0.0695	iShares Short-Term National Muni Bond E...	Municipal Bond	LONG
10	SHM	0.0667	State Street SPDR Nuveen ICE Short Term...	Municipal Bond	LONG

Cluster 17 – Gold

Members: 12 **Var explained:** 0.861 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-12 – filter **PASS** (252d at date = +75.1%)
- Latest sell_cross: 2026-05-20
- Current 252d return: +24.7% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-05-20 more recent than last effective buy 2026-05-12)



Category breakdown

category	n_members	abs_weight
Materials	5	0.4157
Gold	3	0.2502
Precious Metals	2	0.1724
Silver	2	0.1617

Replication basket – GLD+IAU+SLV (cor=0.95) basket lead at lag 1: -0.008

ticker	rep_wt	name	category	etf_signal
GLD	0.022	SPDR Gold Shares	Gold	FLAT
IAU	0.750	iShares Gold Trust Shares	Gold	FLAT
SLV	0.411	iShares Silver Trust	Silver	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
11	DBP	0.0866	Invesco DB Precious Metals Fund	Precious Metals	FLAT
7	GLTR	0.0858	abrdn Physical Precious Metals Basket S...	Precious Metals	FLAT

1	GDX	0.0840	VanEck Gold Miners ETF	Materials	FLAT
2	GDXJ	0.0840	VanEck Junior Gold Miners ETF	Materials	FLAT
3	GLD	0.0834	SPDR Gold Shares	Gold	FLAT
9	IAU	0.0834	iShares Gold Trust Shares	Gold	FLAT
10	SGOL	0.0833	abrtn Physical Gold Shares ETF	Gold	FLAT
12	GOAU	0.0831	US Global GO Gold and Precious Metal Mi...	Materials	FLAT
6	SIL	0.0828	Global X Silver Miners ETF	Materials	FLAT
5	SILJ	0.0818	Amplify Junior Silver Miners ETF	Materials	FLAT
4	SLV	0.0809	iShares Silver Trust	Silver	LONG
8	SIVR	0.0808	abrtn Physical Silver Shares ETF	Silver	LONG

Cluster 18 – Consumer Staples

Members: 11 **Var explained:** 0.744 **Signal:** LONG since 2026-06-15 (6 days, 0.2%)

Signal logic:

- Latest buy_cross: 2026-06-15 – filter **PASS** (252d at date = +7.1%)
- Latest sell_cross: 2026-05-29
- Current 252d return: +7.4% – filter **ALLOWS** new entries today
- Decision: **LONG** (effective buy at 2026-06-15 > last sell at 2026-05-29)



Category breakdown

category	n_members	abs_weight
Consumer Staples	6	0.5733
Utilities	4	0.3519
U.S. Small Cap Consumer Staples	1	0.0747

Replication basket – XLU+XLP (cor=0.99) basket lead at lag 1: -0.010

ticker	rep_wt	name	category	etf_signal
XLU	0.364	State Street Utilities Select Sector ...	Utilities	FLAT
XLP	0.587	State Street Consumer Staples Select ...	Consumer Staples	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
5	VDC	0.0983	Vanguard Consumer Staples ETF	Consumer Staples	LONG
6	FSTA	0.0983	Fidelity MSCI Consumer Staples Index ETF	Consumer Staples	LONG
1	XLP	0.0975	State Street Consumer Staples Select Se...	Consumer Staples	LONG
4	IYK	0.0949	iShares U.S. Consumer Staples ETF	Consumer Staples	LONG
3	KXI	0.0946	iShares Global Consumer Staples ETF	Consumer Staples	FLAT
11	PBJ	0.0897	Invesco Food & Beverage ETF	Consumer Staples	FLAT
7	IDU	0.0890	iShares U.S. Utilities ETF	Utilities	FLAT
9	FUTY	0.0878	Fidelity MSCI Utilities Index ETF	Utilities	FLAT
8	VPU	0.0877	Vanguard Utilities ETF	Utilities	FLAT
2	XLU	0.0874	State Street Utilities Select Sector SP...	Utilities	FLAT
10	PSCC	0.0747	Invesco S&P SmallCap Consumer Staples E...	U.S. Small Cap Consumer ...	LONG

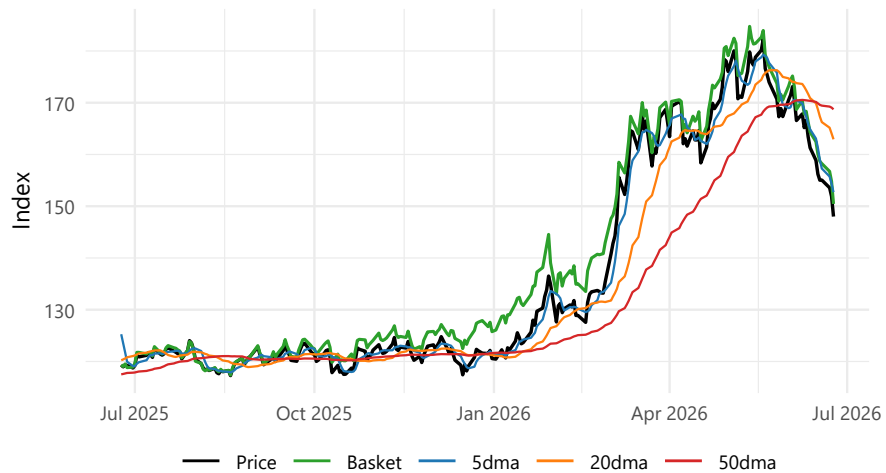
Cluster 19 – Broad Commodities

Members: 11 **Var explained:** 0.772 **Signal:** LONG since 2026-06-03 (14 days, -14.5%)

Signal logic:

- Latest buy_cross: 2026-06-03 – filter **PASS** (252d at date = +48.1%)
- Latest sell_cross: 2026-05-27
- Current 252d return: +20.4% – filter **ALLOWS** new entries today

- Decision: **LONG** (effective buy at 2026-06-03 > last sell at 2026-05-27)



Category breakdown

category	n_members	abs_weight
Broad Commodities	6	0.6172
Oil	2	0.1980
Energy Commodities	1	0.1019
Agricultural Commodities	1	0.0498
Natural Gas	1	0.0330

Replication basket – PDBC (cor=0.98) basket lead at lag 1: 0.079

ticker	rep_wt	name	category	etf_signal
PDBC	1.121	Invesco Optimum Yield Diversified Com...	Broad Commodities	LONG

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 DBC	0.1055	Invesco DB Commodity Index Tracking Fund	Broad Commodities	LONG

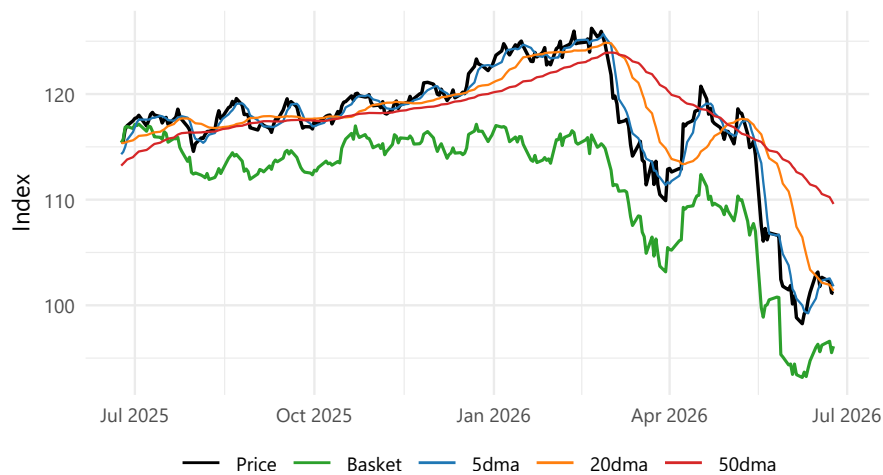
6	GSG	0.1055	iShares GSCI Commodity-Indexed Trust Fu...	Broad Commodities	FLAT
8	PDBC	0.1055	Invesco Optimum Yield Diversified Commo...	Broad Commodities	LONG
9	COMT	0.1052	iShares GSCI Commodity Dynamic Roll Str...	Broad Commodities	FLAT
10	DBE	0.1019	Invesco DB Energy Fund	Energy Commodities	LONG
1	USO	0.0995	United States Oil Fund	Oil	LONG
5	DBO	0.0985	Invesco DB Oil Fund	Oil	FLAT
7	DJP	0.0982	iPath Bloomberg Commodity Index Total R...	Broad Commodities	LONG
4	FTGC	0.0973	First Trust Global Tactical Commodity S...	Broad Commodities	FLAT
3	DBA	0.0498	Invesco DB Agriculture Fund	Agricultural Commodities	FLAT
11	UNG	0.0330	United States Natural Gas Fund LP	Natural Gas	FLAT

Cluster 20 – India Equity

Members: 10 **Var explained:** 0.405 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-06 – filter **PASS** (252d at date = +4.7%)
- Latest sell_cross: 2026-05-13
- Current 252d return: -11.1% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-05-13 more recent than last effective buy 2026-05-06)



Category breakdown

category	n_members	abs_weight
India Equity	3	0.4674
Malaysia Equity	1	0.1126
Indonesia Equity	1	0.1044
Philippines Equity	1	0.1011
Vietnam Equity	1	0.0752
Turkey Equity	1	0.0610
U.S. Dividend Equity	1	0.0548
Platinum	1	0.0234

Replication basket – NOBL+INDA+PPLT (cor=0.91) basket lead at lag 1: -0.007

ticker	rep_wt	name	category	etf_signal
NOBL	0.080	ProShares S&P 500 Dividend Aristocrat...	U.S. Dividend Equity	FLAT
INDA	0.660	iShares MSCI India ETF	India Equity	FLAT
PPLT	0.029	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
1	INDA	0.1572	iShares MSCI India ETF	India Equity	FLAT
2	INDY	0.1555	iShares India 50 ETF	India Equity	FLAT
3	EPI	0.1548	WisdomTree India Earnings Fund	India Equity	FLAT
8	EWM	0.1126	iShares MSCI Malaysia Index Fund	Malaysia Equity	FLAT
6	EIDO	0.1044	iShares MSCI Indonesia ETF	Indonesia Equity	FLAT
7	EPHE	0.1011	iShares MSCI Philippines ETF	Philippines Equity	FLAT
10	VNM	0.0752	VanEck Vietnam ETF	Vietnam Equity	FLAT
9	TUR	0.0610	iShares MSCI Turkey ETF	Turkey Equity	FLAT
4	NOBL	0.0548	ProShares S&P 500 Dividend Aristocrats ...	U.S. Dividend Equity	FLAT
5	PPLT	0.0234	abrdn Physical Platinum Shares ETF	Platinum	FLAT

Cluster 21 – Inflation-Protected Fixed Income

Members: 10 **Var explained:** 0.906 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-05-27 – filter **PASS** (252d at date = +4.8%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +2.6% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2026-05-27)



Category breakdown

category	n_members	abs_weight
Inflation-Protected Fixed Income	10	1

Replication basket – VTIP+SCHP (cor=0.99) basket lead at lag 1: -0.016

ticker	rep_wt	name	category	etf_signal
VTIP	0.262	Vanguard Short-Term Inflation-Protect...	Inflation-Protected F...	FLAT
SCHP	0.762	Schwab U.S. TIPS ETF	Inflation-Protected F...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
4	TDTF	0.1035	FlexShares iBoxx 5 Year Target Duration...	Inflation-Protected Fixe...	LONG

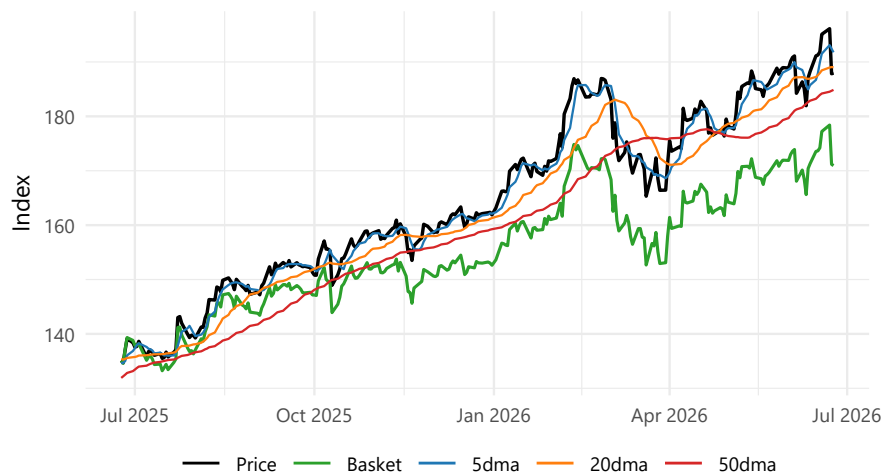
6	TIPX	0.1034	State Street SPDR Bloomberg 1a??10 Year...	Inflation-Protected Fixe...	FLAT
3	TIP	0.1030	iShares TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
8	SCHP	0.1023	Schwab U.S. TIPS ETF	Inflation-Protected Fixe...	FLAT
5	TDDT	0.1007	FlexShares iBoxx 3 Year Target Duration...	Inflation-Protected Fixe...	FLAT
7	SPIP	0.0999	State Street SPDR Portfolio TIPS ETF	Inflation-Protected Fixe...	FLAT
9	STIP	0.0998	iShares 0-5 Year TIPS Bond ETF	Inflation-Protected Fixe...	FLAT
2	STPZ	0.0996	PIMCO 1-5 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT
10	VTIP	0.0994	Vanguard Short-Term Inflation-Protected...	Inflation-Protected Fixe...	FLAT
1	LTPZ	0.0885	Pimco 15 Year U.S. TIPS Index Exchange...	Inflation-Protected Fixe...	FLAT

Cluster 22 – Japan Equity

Members: 9 **Var explained:** 0.888 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-09 – filter **PASS** (252d at date = +57.2%)
- Latest sell_cross: 2026-06-11
- Current 252d return: +39.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-11 more recent than last effective buy 2026-04-09)



Category breakdown

category	n_members	abs_weight
Japan Equity	7	0.7803
Asia-Pacific Equity	2	0.2197

Replication basket – EWJ (cor=0.97) basket lead at lag 1: -0.041

ticker	rep_wt	name	category	etf_signal
EWJ	0.909	iShares MSCI Japan Index Fund	Japan Equity	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

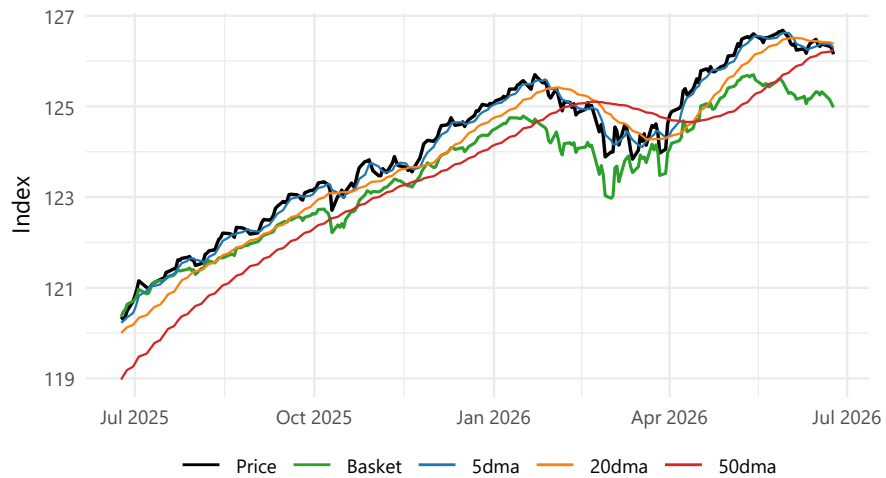
	ticker	PC_wt	name	category	etf_signal
1	EWJ	0.1142	iShares MSCI Japan Index Fund	Japan Equity	FLAT
8	BBJP	0.1141	JPMorgan BetaBuilders Japan ETF	Japan Equity	FLAT
9	HEWJ	0.1117	iShares Currency Hedged MSCI Japan ETF	Japan Equity	LONG
6	DBJP	0.1116	Xtrackers MSCI Japan Hedged Equity ETF	Japan Equity	LONG
3	IPAC	0.1113	iShares Core MSCI Pacific ETF	Asia-Pacific Equity	LONG
5	FLJH	0.1108	Franklin FTSE Japan Hedged ETF	Japan Equity	FLAT
2	DXJ	0.1097	WisdomTree Japan Hedged Equity Fund	Japan Equity	FLAT
4	VPL	0.1084	Vanguard FTSE Pacific ETF	Asia-Pacific Equity	FLAT
7	EWJV	0.1081	iShares MSCI Japan Value ETF	Japan Equity	FLAT

Cluster 23 – Bank Loan / Floating Rate

Members: 9 **Var explained:** 0.558 **Signal:** FLAT

Signal logic:

- Latest buy_cross: 2026-04-07 – filter **PASS** (252d at date = +6.5%)
- Latest sell_cross: 2026-06-23
- Current 252d return: +4.9% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-23 more recent than last effective buy 2026-04-07)



Category breakdown

category	n_members	abs_weight
Bank Loan / Floating Rate	3	0.3627
Interest Rate Hedged Fixed Income	3	0.3245
Floating Rate Fixed Income	3	0.3128

Replication basket – FLOT+BKLN+SRLN+FLRN+FLTR (cor=0.89) basket lead at lag 1: -0.047

ticker	rep_wt	name	category	etf_signal
FLOT	0.293	iShares Floating Rate Bond ETF	Floating Rate Fixed I...	FLAT
BKLN	0.208	Invesco Senior Loan ETF	Bank Loan / Floating ...	FLAT
SRLN	0.310	State Street Blackstone Senior Loan E...	Bank Loan / Floating ...	FLAT
FLRN	0.127	State Street SPDR Bloomberg Investmen...	Floating Rate Fixed I...	FLAT
FLTR	0.035	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
2 SRLN	0.1250	State Street Blackstone Senior Loan ETF	Bank Loan / Floating Rate	FLAT

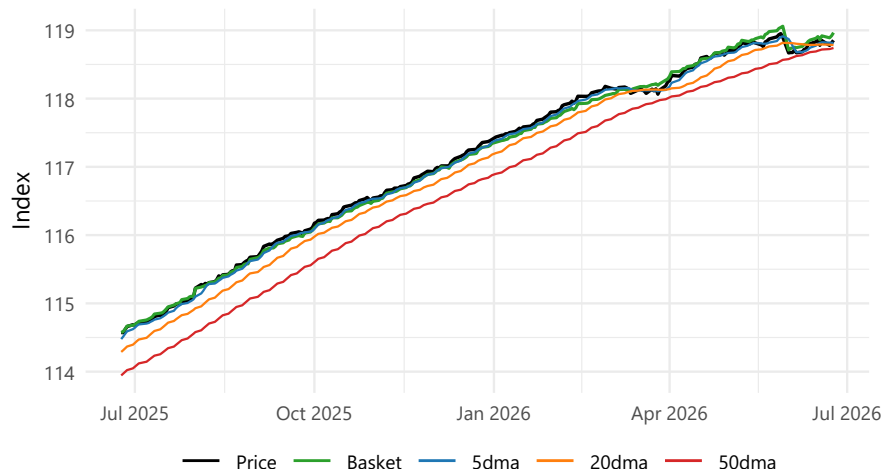
3	BKLN	0.1210	Invesco Senior Loan ETF	Bank Loan / Floating Rate	FLAT
1	FTSL	0.1168	First Trust Senior Loan Fund	Bank Loan / Floating Rate	FLAT
5	LQDH	0.1145	iShares Interest Rate Hedged Corporate ...	Interest Rate Hedged Fix...	FLAT
7	FLOT	0.1082	iShares Floating Rate Bond ETF	Floating Rate Fixed Inco...	FLAT
6	IGBH	0.1057	iShares Interest Rate Hedged Long-Term ...	Interest Rate Hedged Fix...	FLAT
4	IGHG	0.1043	ProShares Investment Grade-Interest Rat...	Interest Rate Hedged Fix...	FLAT
8	FLRN	0.1037	State Street SPDR Bloomberg Investment ...	Floating Rate Fixed Inco...	FLAT
9	FLTR	0.1009	VanEck IG Floating Rate ETF	Floating Rate Fixed Inco...	FLAT

Cluster 24 – Cash / Ultra-Short Fixed Income

Members: 9 Var explained: 0.453 Signal: FLAT

Signal logic:

- Latest buy_cross: 2022-10-28 – filter **BLOCKED** (252d at date = -0.5%; needs > 0)
- Latest effective buy: 2022-11-29 – delayed re-arm from blocked cross (252d at fire = +0.0%)
- Latest sell_cross: 2026-06-03
- Current 252d return: +3.8% – filter **ALLOWS** new entries today
- Decision: **FLAT** (last sell 2026-06-03 more recent than last effective buy 2022-11-29)



Category breakdown

category	n_members	abs_weight
Cash / Ultra-Short Fixed Income	9	1

Replication basket – SHV+MINT+GBIL+ICSH (cor=0.84) basket lead at lag 1: 0.088

ticker	rep_wt	name	category	etf_signal
SHV	0.297	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fi...	FLAT
MINT	0.177	PIMCO Enhanced Short Maturity Active ...	Cash / Ultra-Short Fi...	FLAT
GBIL	0.121	Goldman Sachs Access Treasury 0-1 Yea...	Cash / Ultra-Short Fi...	FLAT
ICSH	0.413	iShares Ultra Short Duration Bond Act...	Cash / Ultra-Short Fi...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
7	SHV	0.1284	iShares 0-1 Year Treasury Bond ETF	Cash / Ultra-Short Fixed...	FLAT
6	ICSH	0.1277	iShares Ultra Short Duration Bond Activ...	Cash / Ultra-Short Fixed...	FLAT
5	GSY	0.1251	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fixed...	FLAT
3	FTSM	0.1204	First Trust Enhanced Short Maturity ETF	Cash / Ultra-Short Fixed...	FLAT
4	NEAR	0.1104	iShares Short Duration Bond Active ETF	Cash / Ultra-Short Fixed...	FLAT
2	MINT	0.1098	PIMCO Enhanced Short Maturity Active Ex...	Cash / Ultra-Short Fixed...	FLAT
1	VNLA	0.0990	Janus Henderson Short Duration Income E...	Cash / Ultra-Short Fixed...	FLAT
8	BIL	0.0949	State Street SPDR Bloomberg 1-3 Month T...	Cash / Ultra-Short Fixed...	FLAT
9	GBIL	0.0843	Goldman Sachs Access Treasury 0-1 Year ...	Cash / Ultra-Short Fixed...	FLAT

Cluster 25 – International Fixed Income

Members: 6 **Var explained:** 0.742 **Signal:** LONG since 2026-05-29 (17 days, -2.6%)

Signal logic:

- Latest buy_cross: 2026-05-29 – filter **PASS** (252d at date = +0.6%)
- Latest sell_cross: 2026-05-18
- Current 252d return: -3.3% – filter **BLOCKS** new entries today
- Decision: **LONG** (effective buy at 2026-05-29 > last sell at 2026-05-18)



Category breakdown

category	n_members	abs_weight
Currency	4	0.6409
International Fixed Income	2	0.3591

Replication basket – BWX (cor=0.93) basket lead at lag 1: 0.026

ticker	rep_wt	name	category	etf_signal
BWX	0.751	State Street SPDR Bloomberg Internati...	International Fixed I...	FLAT

Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

	ticker	PC_wt	name	category	etf_signal
2	IGOV	0.1801	iShares International Treasury Bond ETF	International Fixed Inco...	FLAT
1	BWX	0.1790	State Street SPDR Bloomberg Internation...	International Fixed Inco...	FLAT
4	FXE	0.1702	Invesco CurrencyShares Euro Currency Tr...	Currency	FLAT
5	FXF	0.1649	Invesco CurrencyShares Swiss Franc Trust	Currency	FLAT
3	FXB	0.1610	Invesco CurrencyShares British Pound St...	Currency	FLAT
6	FXY	0.1448	Invesco CurrencyShares Japanese Yen Tru...	Currency	FLAT

Cluster 26 – Leveraged / Inverse**Members:** 2 **Var explained:** 0.625 **Signal:** FLAT**Signal logic:**

- Latest buy_cross: 2026-02-05 – filter **BLOCKED** (252d at date = -32.0%; needs > 0)
- Latest effective buy: 2025-04-08 – delayed re-arm from blocked cross (252d at fire = +0.4%)
- Latest sell_cross: 2026-06-18
- Current 252d return: -37.7% – filter **BLOCKS** new entries today
- Decision: **FLAT** (last sell 2026-06-18 more recent than last effective buy 2025-04-08)

**Category breakdown**

category	n_members	abs_weight
Currency	1	0.5
Leveraged / Inverse	1	0.5

Replication basket – SQQQ+UUP (cor=1.00) basket lead at lag 1: -0.022

ticker	rep_wt	name	category	etf_signal
SQQQ	0.506	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT

UUP	0.495	Invesco DB USD Index Bullish Fund ETF	Currency	LONG
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Members (PC_wt = |PC1 loading|, normalised to unit gross exposure per cluster)

ticker	PC_wt	name	category	etf_signal
SQQQ	0.5	ProShares UltraPro Short QQQ	Leveraged / Inverse	FLAT
UUP	0.5	Invesco DB USD Index Bullish Fund ETF	Currency	LONG

Long ETF Positions (full)

Full reference list of currently-long ETFs. Sorted by cluster's NSS desc, then ETF NSS within the cluster – leading themes appear first, same-theme ETFs are grouped. The actionable buy list is the Position Sheet up front; this table is for cases where you want to inspect all member-level positions.

ticker	cluster	theme	entry	days	return	p_sell	nss
XLF	cluster_14	Financials	2026-04-13	50	4.0%	0.00	1.00
FNCL	cluster_14	Financials	2026-04-13	50	4.0%	0.00	1.00
FXO	cluster_14	Financials	2026-06-05	12	3.8%	0.00	1.00
IAT	cluster_14	Financials	2026-05-20	23	7.8%	0.00	1.00
KIE	cluster_14	Financials	2026-06-10	9	2.8%	0.00	0.98
IAI	cluster_14	Financials	2026-04-10	51	6.7%	0.02	0.98
PSCH	cluster_12	Healthcare	2026-04-13	50	17.5%	0.00	1.00
XBI	cluster_12	Healthcare	2026-06-16	5	10.6%	0.00	1.00
IBB	cluster_12	Healthcare	2026-06-17	4	4.7%	0.00	1.00
PTH	cluster_12	Healthcare	2026-06-17	4	5.4%	0.00	1.00
IHF	cluster_12	Healthcare	2026-05-08	31	7.1%	0.05	0.95
VHT	cluster_12	Healthcare	2026-05-19	24	5.4%	0.29	0.71
FXH	cluster_12	Healthcare	2026-04-10	51	6.1%	0.36	0.63
FHLC	cluster_12	Healthcare	2026-05-19	24	5.3%	0.43	0.57
RSPH	cluster_12	Healthcare	2026-05-21	22	4.9%	0.50	0.50
XHE	cluster_12	Healthcare	2026-06-09	10	-0.5%	0.29	0.43
IYH	cluster_12	Healthcare	2026-05-20	23	4.4%	0.73	0.27
XLV	cluster_12	Healthcare	2026-05-21	22	3.5%	0.74	0.25
PPH	cluster_12	Healthcare	2026-05-21	22	1.9%	0.84	0.03
XHB	cluster_1	U.S. Mid Cap Equity	2026-05-29	17	11.4%	0.00	1.00
PKB	cluster_1	U.S. Mid Cap Equity	2026-06-16	5	3.0%	0.00	1.00
ITB	cluster_1	U.S. Mid Cap Equity	2026-06-09	10	7.3%	0.00	1.00
PSCD	cluster_1	U.S. Mid Cap Equity	2026-05-27	19	7.4%	0.00	1.00
FIW	cluster_1	U.S. Mid Cap Equity	2026-06-16	5	0.5%	0.01	0.99
PEJ	cluster_1	U.S. Mid Cap Equity	2026-05-26	20	7.1%	0.01	0.99
SDVY	cluster_1	U.S. Mid Cap Equity	2026-06-08	11	3.2%	0.00	0.99
SCHA	cluster_1	U.S. Mid Cap Equity	2026-04-09	52	13.2%	0.01	0.99
ITA	cluster_1	U.S. Mid Cap Equity	2026-05-26	20	2.9%	0.02	0.98
REGL	cluster_1	U.S. Mid Cap Equity	2026-06-12	7	0.5%	0.00	0.97
FXR	cluster_1	U.S. Mid Cap Equity	2026-06-12	7	1.1%	0.03	0.95
ISMD	cluster_1	U.S. Mid Cap Equity	2026-04-09	52	14.9%	0.08	0.92
SCHM	cluster_1	U.S. Mid Cap Equity	2026-04-09	52	10.2%	0.08	0.92
XMMO	cluster_1	U.S. Mid Cap Equity	2026-04-06	55	14.5%	0.08	0.92
XRT	cluster_1	U.S. Mid Cap Equity	2026-05-28	18	3.0%	0.09	0.91
IVOG	cluster_1	U.S. Mid Cap Equity	2026-04-09	52	8.8%	0.11	0.89
MDYG	cluster_1	U.S. Mid Cap Equity	2026-04-09	52	8.8%	0.13	0.87
IJK	cluster_1	U.S. Mid Cap Equity	2026-04-09	52	8.8%	0.13	0.87

FXD	cluster_1	U.S. Mid Cap Equity	2026-05-27	19	3.1%	0.19	0.80
IJS	cluster_1	U.S. Mid Cap Equity	2026-04-10	51	9.1%	0.27	0.73
RSPD	cluster_1	U.S. Mid Cap Equity	2026-06-11	8	0.9%	0.26	0.71
PPA	cluster_1	U.S. Mid Cap Equity	2026-05-27	19	-1.9%	0.54	0.39
XAR	cluster_1	U.S. Mid Cap Equity	2026-05-22	21	0.5%	0.58	0.39
FTXR	cluster_1	U.S. Mid Cap Equity	2026-04-17	46	7.5%	0.77	0.22
XMLV	cluster_1	U.S. Mid Cap Equity	2026-06-12	7	-0.3%	0.26	0.05
SMMV	cluster_1	U.S. Mid Cap Equity	2026-06-15	6	0.7%	0.50	-0.43
SHM	cluster_16	Municipal Bond	2026-05-28	18	0.1%	0.00	1.00
HYMB	cluster_16	Municipal Bond	2026-05-26	20	1.1%	0.01	0.99
NYF	cluster_16	Municipal Bond	2026-05-28	18	0.5%	0.02	0.98
FMB	cluster_16	Municipal Bond	2026-05-29	17	0.6%	0.02	0.98
CMF	cluster_16	Municipal Bond	2026-05-28	18	0.5%	0.02	0.98
MUNI	cluster_16	Municipal Bond	2026-05-28	18	0.2%	0.02	0.98
MUB	cluster_16	Municipal Bond	2026-05-28	18	0.4%	0.05	0.95
ITM	cluster_16	Municipal Bond	2026-05-28	18	0.3%	0.05	0.95
VTEB	cluster_16	Municipal Bond	2026-05-28	18	0.4%	0.06	0.94
TFI	cluster_16	Municipal Bond	2026-05-28	18	0.4%	0.06	0.93
PZA	cluster_16	Municipal Bond	2026-05-29	17	0.7%	0.07	0.93
HYD	cluster_16	Municipal Bond	2026-05-22	21	1.5%	0.16	0.84
SUB	cluster_16	Municipal Bond	2026-05-27	19	-0.0%	0.19	0.80
HEWJ	cluster_22	Japan Equity	2026-04-08	53	7.0%	0.00	1.00
DBJP	cluster_22	Japan Equity	2026-04-08	53	5.6%	0.02	0.98
IPAC	cluster_22	Japan Equity	2026-06-12	7	-1.4%	0.29	0.56
EDV	cluster_4	Aggregate Fixed Income	2026-06-12	7	3.2%	0.00	1.00
TLT	cluster_4	Aggregate Fixed Income	2026-06-16	5	1.4%	0.00	1.00
IAGG	cluster_4	Aggregate Fixed Income	2026-05-22	21	1.1%	0.00	1.00
SPTL	cluster_4	Aggregate Fixed Income	2026-06-16	5	1.1%	0.00	1.00
VGLT	cluster_4	Aggregate Fixed Income	2026-06-16	5	1.2%	0.00	1.00
TLH	cluster_4	Aggregate Fixed Income	2026-06-16	5	0.9%	0.00	0.99
BAB	cluster_4	Aggregate Fixed Income	2026-06-16	5	0.2%	0.00	0.99
BLV	cluster_4	Aggregate Fixed Income	2026-06-15	6	1.3%	0.01	0.99
BNDX	cluster_4	Aggregate Fixed Income	2026-05-26	20	0.6%	0.02	0.98
IGLB	cluster_4	Aggregate Fixed Income	2026-06-12	7	0.8%	0.06	0.93
SPLB	cluster_4	Aggregate Fixed Income	2026-06-12	7	0.8%	0.07	0.92
AGGY	cluster_4	Aggregate Fixed Income	2026-06-16	5	0.2%	0.03	0.92
VCLT	cluster_4	Aggregate Fixed Income	2026-06-12	7	0.8%	0.08	0.90
FBND	cluster_4	Aggregate Fixed Income	2026-06-17	4	0.6%	0.04	0.79
LMBS	cluster_4	Aggregate Fixed Income	2026-06-16	5	-0.0%	0.01	0.79
SPMB	cluster_4	Aggregate Fixed Income	2026-06-16	5	0.1%	0.10	0.73
BOND	cluster_4	Aggregate Fixed Income	2026-06-17	4	0.7%	0.08	0.72
IUSB	cluster_4	Aggregate Fixed Income	2026-06-17	4	0.5%	0.12	0.68
LQD	cluster_4	Aggregate Fixed Income	2026-06-16	5	0.3%	0.18	0.68
FLCO	cluster_4	Aggregate Fixed Income	2026-06-16	5	0.2%	0.18	0.66
SPAB	cluster_4	Aggregate Fixed Income	2026-06-24	0	0.0%	0.11	0.66

BND	cluster_4	Aggregate Fixed Income	2026-06-24	0	0.0%	0.13	0.62
AGG	cluster_4	Aggregate Fixed Income	2026-06-24	0	0.0%	0.13	0.62
USIG	cluster_4	Aggregate Fixed Income	2026-06-16	5	0.2%	0.19	0.60
QLTA	cluster_4	Aggregate Fixed Income	2026-06-17	4	0.5%	0.19	0.59
SCHZ	cluster_4	Aggregate Fixed Income	2026-06-24	0	0.0%	0.15	0.57
VMBS	cluster_4	Aggregate Fixed Income	2026-06-24	0	0.0%	0.17	0.53
MBB	cluster_4	Aggregate Fixed Income	2026-06-24	0	0.0%	0.19	0.51
CORP	cluster_4	Aggregate Fixed Income	2026-06-16	5	0.2%	0.22	0.48
GTO	cluster_4	Aggregate Fixed Income	2026-06-16	5	-0.4%	0.32	0.26
LDUR	cluster_4	Aggregate Fixed Income	2026-06-16	5	-0.0%	0.39	-0.19
EWT	cluster_9	Emerging Markets Equity	2026-04-08	53	37.1%	0.07	0.93
EWX	cluster_9	Emerging Markets Equity	2026-06-15	6	-2.3%	0.14	0.66
SCHE	cluster_9	Emerging Markets Equity	2026-06-15	6	-3.2%	0.21	0.55
VWO	cluster_9	Emerging Markets Equity	2026-06-15	6	-3.1%	0.23	0.51
DEM	cluster_9	Emerging Markets Equity	2026-04-09	52	6.2%	0.58	0.37
DGS	cluster_9	Emerging Markets Equity	2026-06-15	6	-3.4%	0.39	0.30
PDP	cluster_6	U.S. Extended Market Eq...	2026-04-08	53	15.1%	0.00	1.00
VFMO	cluster_6	U.S. Extended Market Eq...	2026-04-08	53	14.1%	0.00	1.00
MTUM	cluster_6	U.S. Extended Market Eq...	2026-04-08	53	26.5%	0.00	1.00
VTWG	cluster_6	U.S. Extended Market Eq...	2026-04-13	50	14.9%	0.01	0.99
IWO	cluster_6	U.S. Extended Market Eq...	2026-04-13	50	14.8%	0.01	0.99
FTC	cluster_6	U.S. Extended Market Eq...	2026-04-10	51	14.4%	0.02	0.98
IMCG	cluster_6	U.S. Extended Market Eq...	2026-04-09	52	15.8%	0.03	0.97
RPG	cluster_6	U.S. Extended Market Eq...	2026-04-09	52	19.4%	0.06	0.94
IBUY	cluster_6	U.S. Extended Market Eq...	2026-06-15	6	1.8%	0.04	0.93
SPHB	cluster_6	U.S. Extended Market Eq...	2026-04-10	51	23.5%	0.09	0.91
VOT	cluster_6	U.S. Extended Market Eq...	2026-04-14	49	8.9%	0.10	0.90
IPO	cluster_6	U.S. Extended Market Eq...	2026-04-13	50	22.4%	0.11	0.89
BLCN	cluster_6	U.S. Extended Market Eq...	2026-04-15	48	9.7%	0.46	0.49
IWP	cluster_6	U.S. Extended Market Eq...	2026-04-15	48	4.4%	0.64	0.28
ARKK	cluster_6	U.S. Extended Market Eq...	2026-06-17	4	-2.3%	0.40	0.17
FV	cluster_6	U.S. Extended Market Eq...	2026-04-14	49	13.1%	0.86	0.14
ARKQ	cluster_6	U.S. Extended Market Eq...	2026-06-17	4	-5.5%	0.99	-0.95
PSI	cluster_7	U.S. Large Cap Growth	2026-04-07	54	69.5%	0.00	1.00
SOXX	cluster_7	U.S. Large Cap Growth	2026-04-08	53	62.4%	0.02	0.98
FTXL	cluster_7	U.S. Large Cap Growth	2026-04-07	54	73.5%	0.03	0.97
SMH	cluster_7	U.S. Large Cap Growth	2026-04-08	53	46.3%	0.04	0.96
PTF	cluster_7	U.S. Large Cap Growth	2026-04-07	54	36.1%	0.22	0.78
QTUM	cluster_7	U.S. Large Cap Growth	2026-04-10	51	37.0%	0.29	0.71
QTEC	cluster_7	U.S. Large Cap Growth	2026-04-10	51	40.3%	0.30	0.70
XNTK	cluster_7	U.S. Large Cap Growth	2026-04-10	51	33.0%	0.36	0.64
PSCT	cluster_7	U.S. Large Cap Growth	2026-04-07	54	35.7%	0.64	0.36
XSD	cluster_7	U.S. Large Cap Growth	2026-04-09	52	62.0%	0.65	0.35
FXL	cluster_7	U.S. Large Cap Growth	2026-04-09	52	24.3%	0.72	0.27
QQEW	cluster_7	U.S. Large Cap Growth	2026-04-16	47	13.5%	0.73	0.26

NXTG	cluster_7	U.S. Large Cap Growth	2026-04-08	53	28.0%	0.80	0.18
MGV	cluster_2	U.S. Large Cap Value	2026-04-13	50	8.7%	0.00	1.00
VTV	cluster_2	U.S. Large Cap Value	2026-04-13	50	7.4%	0.00	1.00
SCHV	cluster_2	U.S. Large Cap Value	2026-04-10	51	9.1%	0.00	1.00
AIVL	cluster_2	U.S. Large Cap Value	2026-04-10	51	6.9%	0.02	0.98
FEX	cluster_2	U.S. Large Cap Value	2026-04-09	52	8.7%	0.02	0.98
DIA	cluster_2	U.S. Large Cap Value	2026-04-14	49	6.8%	0.06	0.94
IWD	cluster_2	U.S. Large Cap Value	2026-04-10	51	8.6%	0.06	0.94
IWX	cluster_2	U.S. Large Cap Value	2026-04-10	51	9.1%	0.08	0.92
IMCB	cluster_2	U.S. Large Cap Value	2026-04-10	51	10.4%	0.08	0.92
VONV	cluster_2	U.S. Large Cap Value	2026-04-10	51	8.6%	0.09	0.91
PKW	cluster_2	U.S. Large Cap Value	2026-05-20	23	3.1%	0.10	0.90
VO	cluster_2	U.S. Large Cap Value	2026-04-13	50	6.9%	0.11	0.89
PWV	cluster_2	U.S. Large Cap Value	2026-04-02	56	8.9%	0.11	0.89
IWR	cluster_2	U.S. Large Cap Value	2026-04-10	51	7.9%	0.18	0.82
LVHD	cluster_2	U.S. Large Cap Value	2026-06-10	9	0.8%	0.12	0.81
SPLV	cluster_2	U.S. Large Cap Value	2026-06-11	8	1.2%	0.16	0.50
JVAL	cluster_2	U.S. Large Cap Value	2026-04-10	51	13.0%	0.51	0.48
RSP	cluster_2	U.S. Large Cap Value	2026-04-14	49	5.4%	0.59	0.41
VLUE	cluster_2	U.S. Large Cap Value	2026-04-09	52	28.7%	0.63	0.37
SDY	cluster_2	U.S. Large Cap Value	2026-05-20	23	2.1%	0.57	0.35
PRF	cluster_2	U.S. Large Cap Value	2026-04-09	52	8.2%	0.69	0.30
VIG	cluster_2	U.S. Large Cap Value	2026-04-14	49	4.7%	0.71	0.29
DTD	cluster_2	U.S. Large Cap Value	2026-04-13	50	4.5%	0.75	0.24
FTA	cluster_2	U.S. Large Cap Value	2026-05-21	22	2.9%	0.79	0.20
FNDX	cluster_2	U.S. Large Cap Value	2026-04-09	52	7.1%	0.81	0.17
SPHD	cluster_2	U.S. Large Cap Value	2026-06-08	11	1.4%	0.58	0.13
DLN	cluster_2	U.S. Large Cap Value	2026-04-13	50	4.5%	0.83	0.13
SIZE	cluster_2	U.S. Large Cap Value	2026-04-13	50	5.7%	0.84	0.11
FVD	cluster_2	U.S. Large Cap Value	2026-06-10	9	0.7%	0.48	0.10
RWL	cluster_2	U.S. Large Cap Value	2026-04-10	51	7.6%	0.90	0.09
DGRO	cluster_2	U.S. Large Cap Value	2026-04-13	50	4.1%	0.94	0.06
VOOV	cluster_2	U.S. Large Cap Value	2026-04-14	49	4.1%	0.91	-0.07
LGLV	cluster_2	U.S. Large Cap Value	2026-06-12	7	-0.9%	0.62	-0.45
FTCS	cluster_2	U.S. Large Cap Value	2026-06-10	9	0.3%	0.89	-0.68
PSCC	cluster_18	Consumer Staples	2026-06-10	9	4.2%	0.00	1.00
IYK	cluster_18	Consumer Staples	2026-06-09	10	0.6%	0.58	0.01
VDC	cluster_18	Consumer Staples	2026-06-11	8	-0.7%	0.47	-0.28
FSTA	cluster_18	Consumer Staples	2026-06-11	8	-0.6%	0.57	-0.45
XLP	cluster_18	Consumer Staples	2026-06-10	9	-1.2%	0.73	-0.52
NURE	cluster_15	Real Estate	2026-04-13	50	11.9%	0.29	0.71
INDS	cluster_15	Real Estate	2026-06-08	11	2.6%	0.53	-0.15
RWO	cluster_15	Real Estate	2026-06-09	10	0.4%	0.57	-0.38
PCY	cluster_11	High Yield Bond	2026-05-28	18	0.9%	0.05	0.94
FALN	cluster_11	High Yield Bond	2026-06-12	7	0.3%	0.02	0.93

VVOB	cluster_11	High Yield Bond	2026-05-26	20	1.0%	0.05	0.93
HYEM	cluster_11	High Yield Bond	2026-04-13	50	1.3%	0.05	0.93
ANGL	cluster_11	High Yield Bond	2026-06-15	6	0.0%	0.07	0.75
HYLS	cluster_11	High Yield Bond	2026-06-16	5	-0.1%	0.11	0.13
FPE	cluster_11	High Yield Bond	2026-05-27	19	0.1%	0.06	0.01
HYS	cluster_11	High Yield Bond	2026-06-16	5	-0.1%	0.37	-0.01
JNK	cluster_11	High Yield Bond	2026-06-16	5	-0.2%	0.41	-0.08
SJNK	cluster_11	High Yield Bond	2026-06-17	4	0.0%	0.36	-0.13
SHYG	cluster_11	High Yield Bond	2026-06-17	4	0.1%	0.37	-0.15
BBHY	cluster_11	High Yield Bond	2026-06-16	5	-0.2%	0.44	-0.16
HYG	cluster_11	High Yield Bond	2026-06-17	4	0.2%	0.44	-0.17
HYLB	cluster_11	High Yield Bond	2026-06-17	4	0.1%	0.41	-0.20
UUP	cluster_26	Leveraged / Inverse	2026-05-18	25	3.0%	0.00	1.00
IHDG	cluster_3	Developed ex-US Equity	2026-05-06	33	2.9%	0.00	1.00
EWO	cluster_3	Developed ex-US Equity	2026-04-10	51	9.3%	0.00	1.00
EWN	cluster_3	Developed ex-US Equity	2026-04-10	51	8.3%	0.04	0.96
EWI	cluster_3	Developed ex-US Equity	2026-04-08	53	3.9%	0.16	0.82
EUFN	cluster_3	Developed ex-US Equity	2026-06-12	7	-1.6%	0.11	0.78
GSIE	cluster_3	Developed ex-US Equity	2026-06-12	7	-2.3%	0.34	0.39
DWM	cluster_3	Developed ex-US Equity	2026-06-15	6	-1.6%	0.33	0.33
IQLT	cluster_3	Developed ex-US Equity	2026-06-12	7	-3.0%	0.84	-0.76
SPHQ	cluster_5	U.S. Large Cap Equity	2026-04-10	51	10.5%	0.01	0.99
QUAL	cluster_5	U.S. Large Cap Equity	2026-04-14	49	4.9%	0.94	-0.10
EWS	cluster_8	Asia-Pacific Equity	2026-06-17	4	0.2%	0.13	0.81
EMLC	cluster_8	Asia-Pacific Equity	2026-06-16	5	-1.7%	0.26	0.26
EBND	cluster_8	Asia-Pacific Equity	2026-06-17	4	-1.0%	0.29	0.09
SIVR	cluster_17	Gold	2026-05-27	19	-23.3%	1.00	-1.00
SLV	cluster_17	Gold	2026-05-27	19	-23.3%	1.00	-1.00
ASHR	cluster_13	China Equity	2026-06-15	6	-0.7%	0.02	0.94
CNYA	cluster_13	China Equity	2026-06-15	6	-0.8%	0.03	0.88
TDTF	cluster_21	Inflation-Protected Fix...	2026-05-28	18	-1.9%	1.00	-1.00
XLB	cluster_10	Materials	2026-06-16	5	-3.0%	0.18	0.55
VAW	cluster_10	Materials	2026-06-16	5	-3.4%	0.28	0.34
RSPM	cluster_10	Materials	2026-06-16	5	-3.1%	0.30	0.26
IYM	cluster_10	Materials	2026-06-15	6	-5.2%	0.67	-0.24
MLPX	cluster_10	Materials	2026-06-08	11	-0.2%	0.83	-0.79
MLPA	cluster_10	Materials	2026-06-08	11	-4.2%	1.00	-1.00
DBE	cluster_19	Broad Commodities	2026-06-03	14	-19.0%	1.00	-1.00
USO	cluster_19	Broad Commodities	2026-06-03	14	-24.5%	1.00	-1.00
DJP	cluster_19	Broad Commodities	2026-06-03	14	-12.2%	1.00	-1.00
DBC	cluster_19	Broad Commodities	2026-06-03	14	-12.7%	1.00	-1.00
PDBC	cluster_19	Broad Commodities	2026-06-03	14	-12.6%	1.00	-1.00

ETF Net Signal Strength

NSS = P(buy within h=5 days) - P(sell within h=5 days), range [-1, 1]. ETF rows only, cluster-deduped (top 1 ETF per cluster) so each row is a distinct theme. Bullish side filters to ETFs whose cluster has positive trailing return – i.e. would not be blocked by the entry filter.

Top 20 most bullish (cluster-deduped, abs-mom approved)

ticker	cluster	theme	p_buy	nss
PSCH	cluster_12	Healthcare	1.00	1.00
KBWB	cluster_14	Financials	1.00	1.00
PSCI	cluster_1	U.S. Mid Cap Equity	1.00	1.00
RDVY	cluster_2	U.S. Large Cap Value	1.00	1.00
EDV	cluster_4	Aggregate Fixed Income	1.00	1.00
PSCC	cluster_18	Consumer Staples	1.00	1.00
PDP	cluster_6	U.S. Extended Market Equi...	1.00	1.00
HEZU	cluster_3	Developed ex-US Equity	1.00	1.00
PSI	cluster_7	U.S. Large Cap Growth	1.00	1.00
SHM	cluster_16	Municipal Bond	1.00	1.00
HEWJ	cluster_22	Japan Equity	1.00	1.00
CNXT	cluster_13	China Equity	1.00	0.99
SPHQ	cluster_5	U.S. Large Cap Equity	1.00	0.99
PCY	cluster_11	High Yield Bond	0.99	0.94
EWT	cluster_9	Emerging Markets Equity	1.00	0.93
MINT	cluster_24	Cash / Ultra-Short Fixed ...	0.99	0.93
EWS	cluster_8	Asia-Pacific Equity	0.95	0.81
FLRN	cluster_23	Bank Loan / Floating Rate	0.96	0.72
NURE	cluster_15	Real Estate	1.00	0.71
XLB	cluster_10	Materials	0.74	0.55

Top 20 most bearish (cluster-deduped)

ticker	cluster	theme	p_sell	nss
NORW	cluster_10	Materials	1.00	-1.00
FXC	cluster_8	Asia-Pacific Equity	1.00	-1.00
PDBC	cluster_19	Broad Commodities	1.00	-1.00
IDV	cluster_3	Developed ex-US Equity	1.00	-1.00
XLC	cluster_5	U.S. Large Cap Equity	1.00	-1.00
KGRN	cluster_13	China Equity	1.00	-1.00

ACWV	cluster_2	U.S. Large Cap Value	1.00	-1.00
FXF	cluster_25	International Fixed Income	1.00	-1.00
EWM	cluster_20	India Equity	1.00	-1.00
SLV	cluster_17	Gold	1.00	-1.00
PNQI	cluster_6	U.S. Extended Market Equi...	1.00	-1.00
STPZ	cluster_21	Inflation-Protected Fixed...	1.00	-1.00
PSP	cluster_14	Financials	1.00	-1.00
SDIV	cluster_9	Emerging Markets Equity	1.00	-1.00
PBJ	cluster_18	Consumer Staples	1.00	-1.00
BKLN	cluster_23	Bank Loan / Floating Rate	0.99	-0.99
IWY	cluster_7	U.S. Large Cap Growth	1.00	-0.99
PFXF	cluster_11	High Yield Bond	0.98	-0.97
IHI	cluster_12	Healthcare	0.96	-0.96
REM	cluster_1	U.S. Mid Cap Equity	0.92	-0.92

Diversifier Candidates

Top ETFs ranked by lowest composite of (max correlation against any peer, |PC1 loading|). Both small means the ETF moves uniquely – candidates for portfolio diversification regardless of which cluster they sit in.

ticker	name	category	cluster_id	cluster_size	etf_signal	max_corr_to_others	median_corr_to_others	abs_pc1_loading	compo
PPLT	abrdn Physical Platinum Shares ...	Platinum	cluster_20	10	FLAT	0.305	0.096	0.00021	
UNG	United States Natural Gas Fund ...	Natural Gas	cluster_19	11	FLAT	0.485	0.076	0.00017	
GBIL	Goldman Sachs Access Treasury 0...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.716	0.041	0.00007	
FXV	Invesco CurrencyShares Japanese...	Currency	cluster_25	6	FLAT	0.730	0.095	0.00022	
VNLA	Janus Henderson Short Duration ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.616	0.218	0.00036	
DBA	Invesco DB Agriculture Fund	Agricultural Commodit...	cluster_19	11	FLAT	0.596	0.172	0.00039	
FTSM	First Trust Enhanced Short Matu...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.756	0.135	0.00032	
MINT	PIMCO Enhanced Short Maturity A...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.660	0.186	0.00046	
ICSH	iShares Ultra Short Duration Bo...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.756	0.163	0.00040	
NEAR	iShares Short Duration Bond Act...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.707	0.193	0.00045	
GSY	Invesco Ultra Short Duration ETF	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.734	0.180	0.00043	
BIL	State Street SPDR Bloomberg 1-3...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.851	0.025	0.00000	
SHV	iShares 0-1 Year Treasury Bond ...	Cash / Ultra-Short Fi...	cluster_24	9	FLAT	0.851	0.083	0.00021	
TUR	iShares MSCI Turkey ETF	Turkey Equity	cluster_20	10	FLAT	0.501	0.220	0.00056	
LDUR	PIMCO Enhanced Low Duration Act...	Fixed Income	cluster_4	58	LONG	0.846	0.129	0.00036	
CMBS	iShares CMBS Bond ETF	Mortgage-Backed Fixed...	cluster_4	58	FLAT	0.876	0.161	0.00031	
FXF	Invesco CurrencyShares Swiss Fr...	Currency	cluster_25	6	FLAT	0.820	0.204	0.00049	
LMBS	First Trust Low Duration Opport...	Mortgage-Backed Fixed...	cluster_4	58	LONG	0.837	0.238	0.00047	
NOBL	ProShares S&P 500 Dividend Aris...	U.S. Dividend Equity	cluster_20	10	FLAT	0.445	0.274	0.00075	
SHM	State Street SPDR Nuveen ICE Sh...	Municipal Bond	cluster_16	13	LONG	0.877	0.249	0.00045	
VNM	VanEck Vietnam ETF	Vietnam Equity	cluster_20	10	FLAT	0.444	0.329	0.00083	
SUB	iShares Short-Term National Mun...	Municipal Bond	cluster_16	13	LONG	0.877	0.264	0.00053	
KURE	KraneShares MSCI All China Heal...	China Equity	cluster_13	15	FLAT	0.801	0.271	0.00071	
DBB	Invesco DB Base Metals Fund	Industrial Metals	cluster_10	21	FLAT	0.748	0.362	0.00085	
EIDO	iShares MSCI Indonesia ETF	Indonesia Equity	cluster_20	10	FLAT	0.509	0.360	0.00102	
EPHE	iShares MSCI Philippines ETF	Philippines Equity	cluster_20	10	FLAT	0.564	0.411	0.00107	
FLTR	VanEck IG Floating Rate ETF	Floating Rate Fixed I...	cluster_23	9	FLAT	0.865	0.327	0.00071	
KBA	KraneShares Bosera MSCI China A...	China Equity	cluster_13	15	FLAT	0.857	0.259	0.00073	
CNXT	VanEck ChiNext Innovators ETF	China Equity	cluster_13	15	FLAT	0.904	0.241	0.00064	
LTPZ	Pimco 15 Year U.S. TIPS Index E...	Inflation-Protected F...	cluster_21	10	FLAT	0.918	0.228	0.00057	

Run Parameters

Screen params (signal construction) and backtest params (sizing + strategy filters) shown together. The backtest block is populated by `scripts/04_run_backtest.R` and may be absent on screen-only runs.

	scope	parameter	value
short_ma	screen	short_ma	5
buy_ma	screen	buy_ma	50
sell_ma	screen	sell_ma	20
vol_window	screen	vol_window	20
horizon	screen	horizon	5
imminent_horizon	screen	imminent_horizon	1
cluster_smoothing_window	screen	cluster_smoothing_window	3
sell_cushion_k	screen	sell_cushion_k	0.5
sell_cushion_sd_window	screen	sell_cushion_sd_window	20